# Popularity Prediction and Location Analytics Based on Public Data: Application to Restaurants

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Location is one of the most important strategic decisions for a retail firm. It is a risky high-stakes decision insofar that it involves a large investment, is very difficult to rectify, and affects profits and operations for many years in the future. In this paper we consider the location problem for a retail facility, a consumer-facing storefront such as a restaurant, that provides a service and competes with other establishments. The location decision is difficult for the following two reasons: (i) Forecasts of demand and revenue are difficult to make as the firm has no historical demand data of its own and nor can it obtain competitors' demand, revenue, or cost data. Demand is typically highly variable, dependent on the features and tastes of the population of the catchment area and also what other competing establishments are in the area. (ii) Future entry and exits of competitors would affect demand and the firm's long-term profitability, but predicting such future sporadic events is very difficult. In this paper we show how publicly available demographics and density data can be combined with online review volume and valences to create good forecasts of demand. We then use these forecasts in a simple model of forward-looking competitive entry and exit for predicting locations that will survive over time. Our methodology gives a tractable toolkit to help a firm's decision process. We show that using data-based location decision-making can increase chances of survival (which we define as operating profitably in a steady state of competition). Our estimation results for a specific city also show that customers' sensitivity to distance depends on the price category of the restaurant and review rating sensitivity also varies across restaurant types.

*Key words*: Popularity prediction; retail; location; analytics; online reviews.

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## 1. Introduction

Where to locate a retail outlet is a central problem for retail firms and plays a large role in determining their profitability and viability. It is a decision with long-term implications on operations and profits and yet has to be taken under a great deal of uncertainty about future demand and competitive landscape.

Given its importance to firms, the facility-location problem has been studied intensely in the fields of Operations Research, Computer Science, Location and Regional Science, Information Systems as well as Economics. There are various flavors of the problem in the literature, but broadly can be split into (a) internal—whether the facilities are internal service centers (such as a distribution center serving retail stores), or (b) competitive—whether the clients are exogenous who *choose* to visit one of many competing facilities. The former stream is the more studied, with a primary focus on optimization and approximation algorithms (Fischetti et al. 2017). This paper is mainly concerned with the latter which is significantly more challenging to model, estimate, and optimize.

In contrast to internal facility-location applications, competitive facility-location models bring the estimation task to the fore with competitor strategic behavior as the background canvas: A firm is considering opening one or more retail facilities (stores, restaurants etc.) in a city with incumbent facilities and potential future competition. Demand is from customers who *choose* to patronize a firm based on distance, tastes, reputation, prices, the quality of the service and the available alternatives. Locating a new facility may both expand the size of the demand in that market, as well as take away a portion of the current demand from the incumbent facilities, and crucially, may also influence the future entry and exit decisions of other firms.

The difficulty in predicting future popularity, especially for the restaurant industry that we consider in this paper, is two-fold: First, the firm has no historic own-demand data (as it has not yet started operating), and the demand depends on the tastes and habits of the population in the catchment area, the incumbent alternatives, as well as the design and features and quality of the retail facility. Nor can one hope to obtain demand, cost or revenue information from the competing facilities as it is sensitive private data. Second, future competitor entries and exits are difficult to predict and can adversely affect profits: a highly attractive location from a current demand and competitor location viewpoint may turn out to be unprofitable in the near future as too many firms decide to open nearby, diluting demand. The decision is also high-risk: it requires a large investment and once located, because of the sunk costs, employee travel considerations, and earned customer good-will, is difficult to unwind.

For the first task of demand estimation, firms by and large resort to surveys and market-research studies to assess chances of success, but such studies are expensive time-consuming and not particularly reliable. For the latter task of predicting future competition, economists have proposed many game-theoretic beliefs-based dynamic models, which unfortunately become very complicated, difficult to estimate, and computationally intractable to solve; and besides, are based on strong assumptions on beliefs and strategic calculations (see Prescott and Visscher (1977) and Pancras et al. (2012) that give very valid criticisms of such models).

In this paper we tackle the above challenges in a tractable demand model that we calibrate only on publicly available data (to address the first of the above challenges) and a simple model of entry/exit with the firm aiming to survive profitably in a steady-state (that we define precisely later). Our entry/exit model is amenable to solution by integer programming (IP).

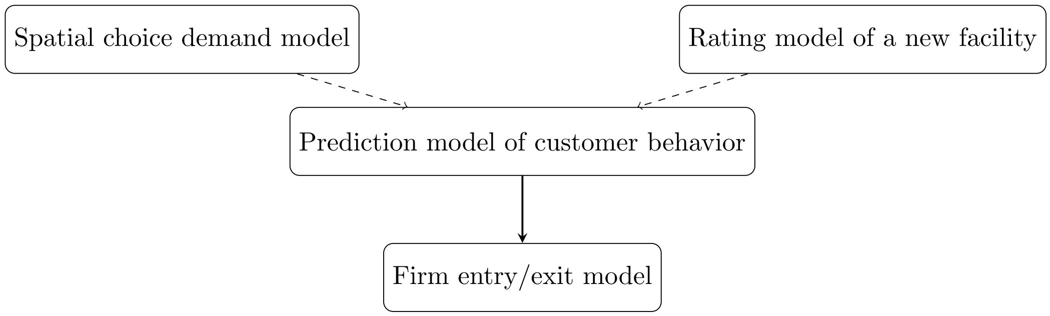


**Figure 1** **Two-stage modeling framework.**

As a specific case-study, we consider the restaurant industry where the econometric analysis is particularly challenging as unobservable factors such as taste, quality and value influence demand. Our contributions on the modeling side for this industry are the following: we show how one can use demographic and density information from public sources and information from online reviews, corrected for endogeneity due to unobservable quality, to make an estimate of what type of facility would gain how much in which location, given an incumbent set of facilities. We show how this demand function can be embedded in a model of sequential entry and exit that is amenable to solution to determine locations that will survive in the long-run. Figure 1 is a schematic of the overall flow of the estimation and optimization models.

In the next subsection, we review the literature on related research. We develop a model of customer behavior in §2 to estimate demand for a facility with particular features at a potential location. In §3 we illustrate numerical results for the estimation applied to the restaurant industry using public data from a specific city. We model firms' entry and exit decision-making in §4. We present our concept of location steady-state in §5; essentially where no firm finds it profitable to enter and incumbent firms are profitable and do not exit. Finally, we conclude with a summary of challenges and managerial insights in §6.

### 1.1. Literature review

Research on facility-location and popularity-prediction problems is being carried out by various research communities.

The early facility-location models in Operations Research deal with internal-serving location decisions such as locating warehouses or factories. The retail outlets are *assigned* to the newly located facilities as part of the solution. This is different from the problem we study in this paper where customers *choose* one of the competing facilities to patronize and the current set of locations influences future competition. We note that a myriad other important variations have been considered: of obnoxious facilities, health or other public services; based on service levels, disruptions etc. (see Lim et al. (2013) for an overview).

A spate of recent articles on location in the Operations Management, Information Systems and Marketing literature raise questions complementary to ours. Pancras et al. (2012) develop a MNL model similar to ours to account for dynamics in goodwill, location endogeneity and spatial competition between retail outlets. They, too, point out the difficulty of obtaining private demand data of all the competing stores even for the researcher, let alone a competing firm. While their analysis highlights the role of endogeneity and location effects, our model is broader in the sense we account for future competition affecting a firm's sales. Srinivasan et al. (2013) is another interesting paper studying the effect of opening and closing of (own) retail stores on a chain's performance. Glaeser et al. (2019) study the location of mobile stores where demand is learned via experimentation.

Notable recent papers in Information Systems that aim to use data to predict popularity are Wang et al. (2022) using machine learning, Li et al. (2021) using location-based social networks such as Foursquare. Our focus on the location decision, notably at a location where the firm has not yet started operations, as well as significantly larger and different data sources, sets us apart from these two papers.

In the Marketing area, Nakanishi and Cooper (1974) is one of the few early papers that addresses the demand estimation for facility location, using a multiplicative competitive-interaction model. Another notable paper on estimation is Ben-Akiva and Watanatada (1981) who develop a continuous spatial-choice multinomial-logit (MNL) model for estimating urban-area travel demand. Irrespective of the exact functional form used, it is common in competitive location models to assume that demand depends on the density and income level of the population as well as the distance to the facility (averaged over the spatial distribution of the population) and attractiveness of the facility.

Relevant to the modeling framework in our paper, Drezner and Drezner (1998), Plastria and Vanhaverbeke (2008), Ghosh and Craig (1983) study *anticipative* competition models in facility-location. They limit themselves to a single competitor however, and permit only entry decisions but not exits. Our modeling approach is quite distinct from these papers as the rate at which entry and exits occur is endogenous in our model. Marianov et al. (2008), Gentile et al. (2018), Dan and Marcotte (2019) solve competitive location by an IP approach as we do in this paper, but do not capture long-term evolution of the competitive landscape. Eiselt et al. (1993), Berman et al. (2009) are a few survey papers specific to competitive facility-location problems.

Pertinent to the restaurant industry, Roy et al. (2022) is a recent article that highlights research opportunities in restaurant analytics. Parsa et al. (2005) present a comprehensive framework from managerial, economic and marketing perspectives on the reasons why so many restaurants fail, and point to location and competitive forces as the main causes. Athey et al. (2018) develop a travel-time factorization model for the restaurant industry and estimate heterogeneous consumer preferences using mobile location data from SafeGraph. A very relevant recent paper Naumzik et al. (2022) addresses survival prediction from online reviews.

Online reviews is a key data source to evaluate satisfaction and have found diverse applications; for instance Chevalier and Mayzlin (2006) argue that consumers use online reviews as a signal of product quality. Luca (2011) finds, using Yelp.com reviews and public restaurant data, that customer reviews affect demand for experience goods. Similarly, Anderson and Magruder (2012) examine the impact of positive rating on demand using Yelp data. Moreover, on the operational side Cui et al. (2017) empirically show that incorporating social media information into sales forecasting improves firms' operations decision-making.

We next turn to "sequential entry" models where firms enter (or exit) one at a time. This contrasts to the simultaneous strategic location decisions as in the classic Hotelling (1929) model; note also that in these papers firms can change location each period, which limits their scope and application to certain types of businesses such as ice-cream trucks on a beach, or mobile stores (Glaeser et al. (2019)).

For a review on sequential location problems, see Eiselt and Laporte (1997). Our entry/exit model is closest in viewpoint to Prescott and Visscher (1977) who construct a dynamic location model anticipating subsequent entrants and exits. Gupta (1992) analyzes sequential entry in a competitive spatial price-discrimination model for both exogenous and endogenous number of firms. Lambertini (2002) analyzes the entry process over an infinite horizon for a duopoly and Aguirregabiria and Vicentini (2016) in a dynamic spatial game in an oligopoly setting.

On the estimation aspect of such sequential location models, Pakes et al. (2007) develop an estimation strategy exploiting information asymmetries in a dynamic model of entry and exit. Our decision model, for our focal firm, has an objective of surviving, and relevant to this, Agarwal and Gort (1996) study evolution of markets and analyze entry, exit and the survival of firms. Amir and Lambson (2003) study survival in a game-theoretic setting in an infinite-horizon model and construct an equilibrium framework where the last firm to enter is the first to exit.

Notably, all the above-mentioned papers on entry and exit modeling assume that the number of players is known. In practice, especially in our setting of small independent retail firms such as cafes and restaurants, this may not be the case. As a general comment on the literature, there is no consensus on the right model for the strategic aspects of entry, nor an operational, estimable and tractable model for entry and exits, as this quote from the survey article of Doraszelski and Pakes (2007) makes clear:

> *"This proliferation of entry models is largely because there is no agreement, and very little in the way of empirical guidance, on the appropriate way to model entry."*

Our entry/exit model, given this state of affairs, abstracts away some aspects of the problem, avoiding onerous assumptions on common beliefs and information, knowledge of all potential entrants and their strategic behavior etc., making it more amenable to estimation, analysis and computational tractability.

## 2. Demand Model

In this section, we describe our spatial-choice model of demand and a model for ratings to predict popularity of a hypothetical establishment at a location and then apply it to the case of our focal industry, restaurants.

### 2.1. Spatial-choice demand model with online reviews information

Potential customers choose one of the open facilities to visit, or decide not to patronize any. Their choice of facility to visit depends on their demographics (e.g. income, age), idiosyncratic tastes, the distance between their residence and of the locations, and finally the price, features and quality reputations of the various alternatives.

Customers are assumed to visit at most one facility on a single trip. This is a reasonable assumption for businesses such as supermarkets or restaurants. Hence discrete-choice models are appropriate, and amongst these we adopt the simplest one, the MNL choice model. This model is frequently used in the facility-location literature (De Palma et al. (1989), Benati (1999), Marianov et al. (2008), Haase and Müller (2014)). Certainly more complicated "realistic" models than MNL can be constructed, but as we shall see even the MNL model captures important aspects of the problem, and already leads to challenging estimation and optimization difficulties due to unobservable factors.

We denote the incumbent set of facilities by $\mathcal{I}$ and the set of available potential locations as $\mathcal{P}$. Each facility is identified by location, and observable as well as latent features that influence the attractiveness of the facility.

We focus on customers who live in a region (a city) and assume we know the population density distribution. The region is divided into many grid points, say by a regular partition of the city area, and we denote the set of grid points by $\mathcal{G}$. We calculate customer-to-facility Euclidean distance from the center of the grid point to the facility. The choice of Euclidean distance is motivated by the rationale that a consumer's indifference curves in geographic space form concentric circles around their location, as discussed in Davis (2006).

The utility of visiting a service facility $i \in \mathcal{I}$ for a customer located in $g \in \mathcal{G}$ is given by

$$U_{ig} = W_{ig} + \upsilon_{ig},$$

where $W_{ig}$ is the deterministic utility and $\upsilon_{ig}$ is a random component representing unobservable factors, that we assume to be i.i.d with a standard Gumbel distribution. The deterministic part of the utility, $W_{ig}$, is given by

$$W_{ig} = \beta_p p_i + \boldsymbol{\beta}_{\boldsymbol{c}}^T c_i + \beta_d d_{i,g} + \beta_r R_i,$$

where $p_i$ denotes price point, $c_i$ is a 0-1 indicator value (dummy) of the cuisine of $i$, $d_{i,g}$ stands for the distance between the location of the facility $i$ and a customer located in grid point $g$, and $R_i$ is a rating of the facility by previous patrons that is a result of price (value) and unobservable quality and idiosyncratic tastes; $\beta_p$, $\beta_d$ and $\beta_r$ are parameters that measure the effect of price category, distance and rating respectively, and $\beta_c$ of the cuisine type. We keep the model concise to limit the computational burden and remind the reader that it can be made as complex or parsimonious as one chooses. For instance, in §3.4, we allow the distance and price coefficients to vary across geography. This leads us to gain insights on how the restaurant characteristics interact with demographical traits by neighbourhoods.

In §2.2, we develop a functional form, Equation (3), for $R_i$, as a function of the attributes and local demographics and latent factors to capture the unobservable quality and idiosyncratic tastes that go into the rating.

Given our MNL model and distributional assumption on the error component $\upsilon_{ig}$, the probability that a customer located at grid point $g$ chooses facility $i$ is given by:

$$P_{ig}(\mathcal{I}) = \frac{e^{W_{ig}}}{1 + \sum_{\imath \in \mathcal{I}} e^{W_{\imath g}}}. \tag{1}$$

Let $M_g$ denote the potential market size in grid point $g$, modeled as $M_g = C_g \left(\frac{E_g}{E_{\max}}\right)^{\beta_z}$, where $C_g$ is the population density of grid point $g$, $E_g$ denotes average income of the population of $g$, $E_{\max}$ is an upper bound on income and $\beta_z$ is a parameter common across all income ranges and grid points to be estimated. Our modeling device of using income-adjusted density is similar to Berman et al. (2009) who call it *buying-power*, a concept well-established in the facility-location literature. The thinking behind this is that population count per se is not representative of the market size as wealth plays a role in the frequency of visiting a service facility (Cowan 2017, Aug 20). Then, the demand at any facility $i$, $\hat{D}_i$, is the sum of the demands from all the grid points,

$$D_i = \sum_{g \in \mathcal{G}} M_g P_{ig}(\mathcal{I}).$$

Success of a service business like a hair saloon, cafe or restaurant depends also on its quality and consequentially, reputation; we consider customer ratings given by online reviews as a proxy that captures both quality, reputation as well as tastes (See-To and Ngai (2018), Chong et al. (2017)).

We use both the volume and valence of reviews in our demand estimation. Specifically, we consider the average review volume $n_i$ for facility $i$ as proportional to the demand for the restaurant multiplied by a fraction who write reviews, and this fraction depends on the price-point of the restaurant; namely:

$$n_i = \mu q_i \hat{D}_i + \varepsilon_i, \tag{2}$$

where the fraction that leave a review is modeled as

$$q_i = \frac{e^{\beta_p^{rev} p_i}}{1 + e^{\beta_p^{rev} p_i}},$$

and $\mu$ is a scaling constant common across all restaurants and $\varepsilon_i$ is an error term.

We wish to remark that lacking data on the observed demand for all the facilities, it is impossible to estimate parameters such as $\mu$ in Equation (2). Our assumption of a common $\mu$ however allows us to eliminate it by taking ratios across restaurants so that the parameter $\mu$ drops out. We subsequently use samples of secondary sources to calibrate and validate the model (§3.3 and §3.4; results in Table 3).

We next describe our structural model for popularity based on latent tastes for a restaurant: how the population of a particular zip-code would evaluate a particular restaurant type. This serves as an input for the demand estimation model described in this section, which in turn drives the optimization model in §4 with future entries and exits.

## 2.2. Latent Factor Model (LFM) to predict the ratings of a new restaurant

It is difficult enough to predict customer tastes and demand for an existing restaurant, so it is not surprising that it is considerably more challenging to do so for a hypothetical future establishment. Tastes, quality, design features, a general vibe of the place so to speak, would influence its success, along with the demographics of its catchment area and the competing alternatives. Our model of ratings for a new restaurant at a location is based on observed online review ratings for the incumbent establishments and the local demographics and a set of latent factors to capture unobservable quality and tastes.

In our model, each restaurant $i$ of cuisine-type $c$ has some restaurant-specific features (e.g: parking available, romantic, serving groups etc.) that are captured in $\boldsymbol{y}$, and a price-point denoted by $p$ (a categorical variable to represent the price category such as $under 10; $$=11-30; $$$=31-60; $$$$-above 60) . Our dependent variable is the mean rating of a facility, the independent variables in the regression are the features of the facility and the location demographics and a set of latent factors for the location and the cuisine-type. The latent factors are intended to capture the unobservable quality of the restaurant as well as the idiosyncratic tastes of the local population.

The model thus is given by:

$$R_i \sim \zeta_0 + \boldsymbol{\zeta_y}^T \boldsymbol{y_i} + \zeta_p p_i + \zeta_c c_i + \boldsymbol{u}_{c_i}^T \boldsymbol{v}_{z_i} + \zeta_a a_i + \zeta_\aleph \aleph_i, \tag{3}$$

where the parameter $\zeta_0$ is an intercept, $\boldsymbol{\zeta_y}$ is a vector of parameters associated with the features, $\zeta_c$ is the coefficient of cuisine-type, $\zeta_p$ is the price coefficient, $\zeta_a$ is the coefficient for restaurant's age, $\zeta_\aleph$ is the coefficient for the number of competitors in the vicinity and finally an inner product of two latent factor vectors $\boldsymbol{u}_{c_i}$ associated with the cuisine of $i$, $c_i$, and $\boldsymbol{v}_{z_i}$ associated with the zip-code $z_i$ of the restaurant $i$. Note that $\boldsymbol{\zeta_y}$ is common to all facility types. We fix the dimension of the latent factors to be 2 based on tuning on a test sample (the errors on test set does not decrease with larger dimensions).

The estimated parameters minimize a loss function, the regularized squared error on the set of observed ratings for the incumbent facilities

$$\min_{\zeta_0, \boldsymbol{\zeta}_y, \zeta_c, \zeta_p, \boldsymbol{u}_{c_i}, \boldsymbol{v}_{z_i}, \zeta_a, \zeta_\aleph} \sum_{i \in \kappa} (R_i - \zeta_0 - \boldsymbol{\zeta_y}^T \boldsymbol{y_i} - \zeta_c c_i - \zeta_p p_i - \boldsymbol{u}_{c_i}^T \boldsymbol{v}_{z_i} + \zeta_a a_i + \zeta_\aleph \aleph_i)^2 + \underbrace{\lambda(||\boldsymbol{u}_{c_i}||^2 + ||\boldsymbol{v}_{z_i}||^2 + ||\zeta_{c_i}||^2)}_{\text{Regularization}},$$

where $\kappa$ is the training set for which $R_i$ is known, $\lambda$ is a regularization term to avoid overfitting (after some tuning, we use regularization value of $\lambda = 0.1$). We remark that ratings is a continuous variable, i.e., can be fractional such as 4.5 stars.

The regression problem is not convex in general, but the alternating gradient-descent method performs well in our case, as is the experience in other settings (Koren et al. 2009). A complication however, that is not addressed in the recommender systems literature, that we observe for restaurants and resolve, is endogeneity. We describe this in §3.2.

## 3. Estimation: Taking the model to data

Our focal industry, the restaurant business, tends to be very local and both location and competition play a large role in the success of a restaurant. The industry has a few large chains, mostly fast-food chains operating at the lower end, but is predominantly made up of small independent businesses.

We describe the data, the estimation of the model, and the results of the popularity prediction next and subsequently the entry/exit prediction and the limiting-state in §5.3.

### 3.1. Public data

Public data is immensely useful as little effort has to be made in collecting and cleaning it and is easily accessible to a small business. We briefly go over the sources we use in our illustrative case study for locating a restaurant.

We use data from Yelp.com's public Dataset Challenge to obtain restaurant information and customer review ratings. Since 2006, Yelp Dataset Challenge has been running annually in which

Yelp posts its reviews data in an easily accessible format online to encourage research and gather insights. The reviews dataset contains business features (e.g. cuisine type, parking, delivery, classification of price points, location (latitude/longitude coordinates, zip code) and customer reviews from which we can gather the number of reviews and their distribution.

We pick a representative city, Las Vegas (Clark County) U.S., to illustrate our results. This city has $3,987$ restaurants located over 39 zip codes, serving 31 types of cuisines, and there are a total of $475,917$ customer reviews for these in our Yelp dataset.

**Residential Population Density:** We obtain demographic information, including average income by zip-code, from the U.S. Census Bureau. We retrieve population density data from NASA (2016) which is available at a very fine grid level. The grid point area dimensions are 0.927 km $\times$ 0.752 km and there are a total of 1526 grid points in our representative city.

**Tourist Volume:** In dense tourism-oriented cities such as Las Vegas, a significant percentage of demand in certain locations consists of transient tourist population. This does not show up in demographic population data and we have to assemble this from other sources. Data from Factual (2016) gives exact geo-location of hotels along with their capacity. Moreover, according to publicly available statistics from Statista (2016), the hotels in this city have an average occupancy of 63% and stay length of 3.5 days. Given these parameters we calculate transient population approximately. Local population for the considered region is nearly 1.4 million and after accounting for transient population available at any given time it increases to 1.76 million, averaged over a year.

**Rental Cost:** We collect approximate rental costs information from Craigslist (2016) and compile price per square foot by zip codes.

**Fixed and Operating Costs:** This can vary by large margins depending on location and decor. To be representative we used the workbook from the accounting software firm Sage (`https://www.sage.com/en-us/-/media/images/sagedotcom/us/en-us/startup-costs/calculators/restaurantcalculatorexcel.xlsx`). There are many other restaurant blogs and websites that corroborate the numbers that come out of this workbook.

**Auxiliary Information:** In addition to the sources mentioned above, we use a few other sources such as OpenTable and Google indirectly and *only* on a small sample, to calibrate and validate our estimation procedures. For instance, we use OpenTable (2016) and Google (2016) for validating our demand estimation model.

For all these sources, we manually collect the necessary data conforming to the websites' terms and conditions. Summary statistics of the data are provided on Table EC.9 in Appendix EC.4.

### 3.2. Rating Prediction: Problem of Endogeneity

A direct application of the model leads to wrong signs on the price coefficient and other biased estimates because of endogeneity issues—unobservable quality influences the co-variates such as price and ratings. Although endogeneity is a well-known issue in many econometric studies, there is relatively little research in the context of the effect of price on ratings or estimating price-sensitivity from reviews data (Dubois and Nauges (2010) being one of the few). We first show evidence of potential endogeneity in the reviews data.

We organize the ratings data from Yelp in a matrix of 39 zip codes and 31 cuisine types to set up the matrix factorization framework of LFM estimation. Each cell in the matrix can have multiple restaurants and a rating is in fact a distribution given by the fraction rating of the facility as 1-Star, 2-Star etc. Each restaurant also has additional features, the so-called "side-information" in LFM, that in our sample case is any combination of (a) groups (b) takes-reservation option, and (c) not including any features. We experimented with each of the explanatory variable (restaurant features) of ratings one at a time and compared the errors to arrive at a decision on what features to include in the model. Although Yelp provides ranges for each price category, we treat price as a categorical variable as the ranges could be changing over time. The total number of unique zip-cuisine pairs in our restaurant dataset is 1625. So, sparsity level is $1 - \frac{1625}{39 \cdot 31 \cdot 4 \cdot 4} = 91.6\%$. We use Root Mean Squared Error which is defined as $\text{RMSE} = \sqrt{\sum_{i=1}^{\mathcal{I}} (R_i - \hat{R}_i)^2 / |\mathcal{I}|}$, to evaluate accuracy of predicted ratings.

We briefly describe the out-of-sample performance for our restaurant data. We compare the means of actual and predicted ratings given by (3) aggregated across a sample of cuisines for a specific zip code region (Table EC.5 on Appendix §EC.3). We also test rating model predictions with and without side information. Overall the LFM performs quite well and we observe relatively low RMSE after convergence. The errors are slightly lower once we include features, and adding features into the model also enhances our knowledge about each features' contribution on ratings (Tables EC.5 and EC.6 in Appendix §EC.3).

In the first line of Table 1, we have the Least Squares (LS) estimation coefficients of different features along with RMSE on both training and test samples. As can be seen, the coefficient of price is estimated to be 0.005, a wrong sign. One naturally suspects endogeneity effects are at play, as an intangible "quality" or a "cool factor" may be behind the higher price. We confirm this in the next section using diagnostic econometric tests.

Table 1 also shows the IV-corrected estimates; having additional features such as serving groups, taking reservations positively affect the ratings; age of the restaurant negatively affects ratings whereas there is slight contribution to ratings as number of competitors increases.

| Method | Constant | Serving groups | Takes reservations | Restaurant age | #comp. $\aleph$ | Price range | RMSE Train (Test) |
|---|---|---|---|---|---|---|---|
| **LS procedure** | $\zeta_0 = 3.069$ | $\zeta_{y_1} = 0.270$ | $\zeta_{y_2} = 0.113$ | $\zeta_a = -0.094$ | $\zeta_\aleph = 0.004$ | $\zeta_p = 0.005$ | 0.635 (0.661) |
| **IV correction** | $\zeta_0 = 3.19$ | $\zeta_{y_1} = 0.268$ | $\zeta_{y_2} = 0.157$ | $\zeta_a = -0.091$ | $\zeta_\aleph = 0.004$ | $\zeta_p = -0.08$ | 0.634 (0.662) |

*Note.* All the variables except for the dummy variables are standardized.

**Table 1 Estimates of rating model based on LFM without and with instrumental variable (IV) correction.**

**3.2.1. Endogeneity correction for unobservable quality** The quality of the facility is unobservable to the researcher but nevertheless affects the consumer's choice via the rating of the facility and potentially, makes the restaurant charge a higher (or lower) price. The price information is only available in $ terms, called as price point. When we omit unobservable quality, it gets absorbed in the error term in (3) and in this case, LS regression leads to a biased $\zeta_p$, an endogeneity problem—regressors being correlated with the error term because of omitted variables.

Instrumental variable (IV) regression—with suitable instruments—is the standard econometric technique to account for omitted variables and obtain consistent estimates of the parameters.Guevara and Ben-Akiva (2006) study a problem similar to ours, namely endogeneity in a residential location-choice model and suggest an instrument proposed by Hausman (1996) and Nevo (2001). As in their study, we use observed average price points (denoted by $\bar{p}_i$) of the same cuisine in adjacent zip-codes as an instrument for price. This is considered a valid IV as the average price point of the same cuisine is correlated with a focal restaurant's price point due to competitor effects and customer expectations but the average price point in neighboring zip-codes is not expected to be correlated with the idiosyncratic quality of the restaurant.

As LFM is a nonlinear model, we use the control-function method of Wooldridge (2015) for the IV estimation. This approach consists of two steps.

*First step:* We use LS regression to estimate the linear projection of the endogenous explanatory variable, specifically price, on the instruments and the rest of exogenous variables, the binary coded features of the restaurant $y_i$ and cuisine dummies, $c_i$.

$$p_i = \theta_0 + \boldsymbol{\theta}_{\boldsymbol{y}}^T y_i + \theta_c c_i + \theta_{\bar{p}} \bar{p}_i + \theta_a a_i + \theta_\aleph \aleph_i + \delta.$$

Using the estimated coefficients in the first step, we obtain estimated values, $\hat{p}_i$ to be used next.

*Second step:* We introduce residuals estimated from the first stage; the residuals being defined as $\hat{\delta}_i = p_i - \hat{p}_i$. Then, we run the following regression for the second step of LFM

$$R_i = \zeta_0 + \beta_{\hat{\delta}} \hat{\delta}_i + \boldsymbol{\zeta}_{\boldsymbol{y}}^T \boldsymbol{y_i} + \zeta_c c_i + \zeta_p p_i + \boldsymbol{u}_{c_i}^T \boldsymbol{v}_{z_i} + + \zeta_a a_i + \zeta_\aleph \aleph_i + \varepsilon_i,$$

where $\beta_{\hat{\delta}}$ is the estimated coefficient for the first step regression residuals.

After correcting for bias using IV, we get a reasonable coefficient estimate for price point, shown in the "IV correction" line of Table 1—notably the price point coefficient is now negative. Specifically, for a restaurant with an average rating of 3.5, if we increase the restaurant's price point by

**Restaurants, sample size of 3987**

| Diagnostics | Statistic | p-value |
|---|---|---|
| Weak instruments | 3201.85 | $\approx 0^{***}$ |
| Wu-Hausman | 18.14 | $\approx 0^{***}$ |

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $.p < 0.1$.

**Table 2 Diagnostic tests for endogeneity.**

one \$ sign, assuming all other factors remain constant, the average rating of the restaurant would decrease to $3.5 - 0.08 \cdot 1 = 3.42$, because of value considerations, which seems plausible.

We also perform diagnostic tests (Baum et al. 2003) to validate the choice of instruments in our study (Table 2). According to the weak-instruments test, we reject the null as p-value $\approx 0$ to conclude that our instruments are not weak. In the case of the Hausman test, we again reject the null as p-value $\approx 0$, so we can claim that estimates obtained using IV correction are not similar to the ones obtained with standard procedure and endogeneity was indeed a problem in our dataset.

### 3.3. Demand Estimation

In this section, we explain how we calibrate the demand model using secondary sources on bookings. As emphasized in §2, we do not observe demand data of the competitors' restaurants and our premise is that customer review *volume* gives us useful information about the demand. Once we assume the number of customers who leave feedback is proportional to demand, we can at least get a share of the total patrons of all the facilities and obtain

$$\frac{n_i}{n} = \frac{q_i D_i}{\sum_{\imath \in \mathcal{I}} q_\imath D_\imath}, \tag{4}$$

where the constant term $\mu$ drops out and using (4) we run a nonlinear least squares regression to estimate the choice parameters, the $\beta$'s. Recall that we cannot pinpoint the number of customers who leaves a review either that we are modeling with $q_i$.

For the restaurant industry, we have bookings data published by a reservation site called OpenTable. From this website, we collect a sample of booking data and verify that the correlation between number of reviews and demand is quite high (Figure 2).

As evident from Figure 2, there is a discernible linear relationship between OpenTable bookings and review volume. We present the figures by price point, highlighting variations in slopes that correspond to each price point. OpenTable is a platform where mostly high-end restaurants exist, so we group low-end restaurants all together under price range of 2. We are aware that the observed demand at the OpenTable platform by itself is not complete as reservations can be made by phone-calls or customers walk in without a reservation. Yet, this is good evidence corroborating our proportionality assumption across restaurants.

Another potential source to infer demand data is Google with its *Popular Times* feature. Since accessing the data is restricted, we use it only to validate the results of our demand model on a randomly collected sample of restaurants and conduct robustness checks in §3.4.1.

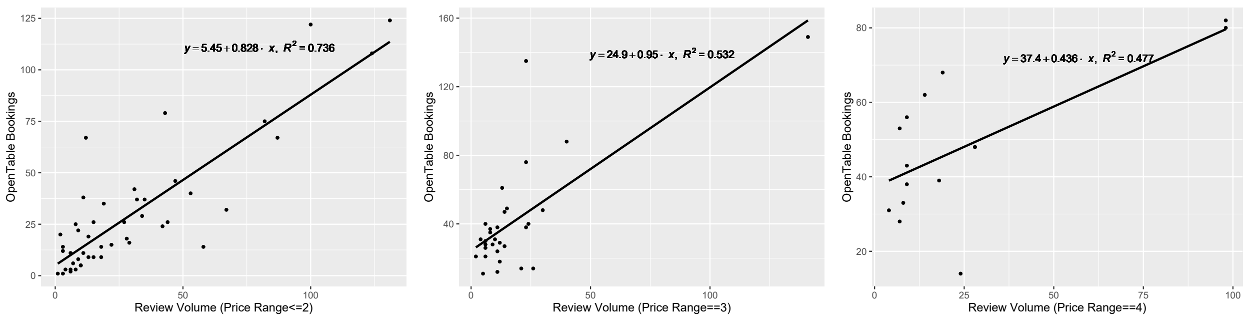


**Figure 2** **The relation between Yelp reviews and OpenTable bookings for a random sample of 92 restaurants by price ranges.**

## 3.4. Model results

We now present the results from our demand model estimation. We preprocess the data by transforming the distance variable to standardize its range between 0 and 20 miles using min-max normalization. We normalize the rating variable also by subtracting the mean and dividing by the standard deviation. We denote the coefficient of rating for the normalized case by $\bar{\beta}_r$.

| **Parameters** | **Market Share** $n_i/n$ | **Elasticities across restaurant types** \$ | \$\$ | \$\$\$(\$) |
|---|---|---|---|---|
| $\beta_r$ *(Std. Avg. Rating)* | $0.503^{***}$ | $0.693^{***}$ | $0.495^{***}$ | 0.166 |
| | (0.043) | (0.085) | (0.051) | (0.083) |
| $\beta_d$ *(Distance)* | $-0.186^{**}$ | $-0.429^{***}$ | $-0.15^{*}$ | -0.148 |
| | (0.068) | (0.164) | (0.065) | (0.321) |
| $\beta_z$ *(Income)* | $1.537^{\cdot}$ | | | |
| | (1.032) | | | |
| $\beta_p$ *(Price)* | $-0.152^{**}$ | | | |
| | (0.04) | | | |
| $\beta_p^{\text{rev}}$ *(Review Fraction)* | $1.686^{**}$ | | | |
| | (0.142) | | | |
| Observations | 3987 | 1872 | 1831 | 284 |

*Note.* Robust standard errors are given in parentheses. $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

**Table 3** **Estimates of demand model for Las Vegas city.**

The results are presented in Table 3 along with robust standard errors. Our elasticity estimates are broadly in agreement with the ones reported in the literature (Pancras et al. (2012)). We also gain insight into the trade-off between increasing review ratings (quality) versus decreasing distance to the customers: our study finds that the first factor has a higher impact. Restaurateurs who do not have sufficient resources to locate in central regions have to compensate by increasing customer satisfaction. As a further check, we group the restaurants by price-ranges and estimate the coefficients of interest separately. We illustrate how distance and rating elasticities vary across restaurant types. We find that rating factor plays a rather important role for inexpensive restaurants (elasticity of 0.693 for price point \$ versus 0.166 for price point \$\$\$(\$)). We also observe that willingness to travel is lower for low-priced restaurants (elasticity of -0.429 for price point \$ versus and -0.148 for price point \$\$\$(\$); Athey et al. (2018) state a similar result). This indicates that

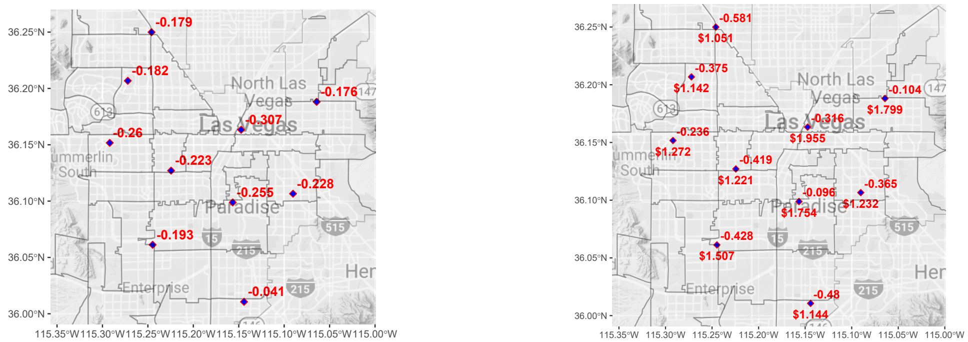


**Figure 3** **Average distance and price elasticity across 10 neighbourhoods in Las Vegas. The numbers in $ signs represent average cost per sq feet of the rentals in each neighbourhood.**

customers of mid-priced restaurants are more sensitive to review ratings compared to those visiting high-priced establishments. One possible explanation is that customers rely more on restaurant critics in newspapers or magazines for the destination restaurants. In contrast, sensitivity to distance is nearly three times lower for high-end restaurants, which is again quite plausible; this would also mean a firm thinking of opening a high-end restaurant has wider options for location.

There is considerable heterogeneity in how distance is viewed across the city. We divide Las Vegas city into 10 neighbourhoods taking into account the city's unique location characteristics. Detailed names and locations can be found at https://www.amandaboltonrealtor.com/community/spring-valley. We exclude Green Valley & Henderson due to the limited number of restaurants in these areas, as reflected in the Yelp data. Then, we measure each of the neighbourhoods' elasticity of distance. Figure 3 illustrates how elasticities vary across several neighbourhoods in the city. We find that in the suburban locations, people are willing-to-travel up to 7.48 times longer distance compared to the city center locations ($\beta_d^{\text{Downtown}} = -0.307$ versus $\beta_d^{\text{Anthem}} = -0.041$). We apply a similar heterogeneity analysis to the price coefficient. Our findings highlight that residents in suburban areas are nearly 6.05 times more sensitive to price compared to those in central locations ($\beta_d^{\text{Centennial}} = -0.581$ versus $\beta_d^{\text{Southeast}} = -0.096$). Moreover, we observe a strong correlation, nearly 0.7, between rental prices and price sensitivities of the neighbourhoods.

**3.4.1. Validation on a hold-out sample** We corroborate the results of the ratings and demand model forecasts using hold-out samples and auxiliary sources. In addition, we perform robustness checks partitioning the data sample to several groups to conform our intuition of estimates.

We gather seat information for a small subset of restaurants from the Building and Safety Department of Nevada (Las Vegas City Hall 2014). We also obtain the *occupancy* percentages and average time spent using *Popular Times* feature of Google Maps at a restaurant level. We cross

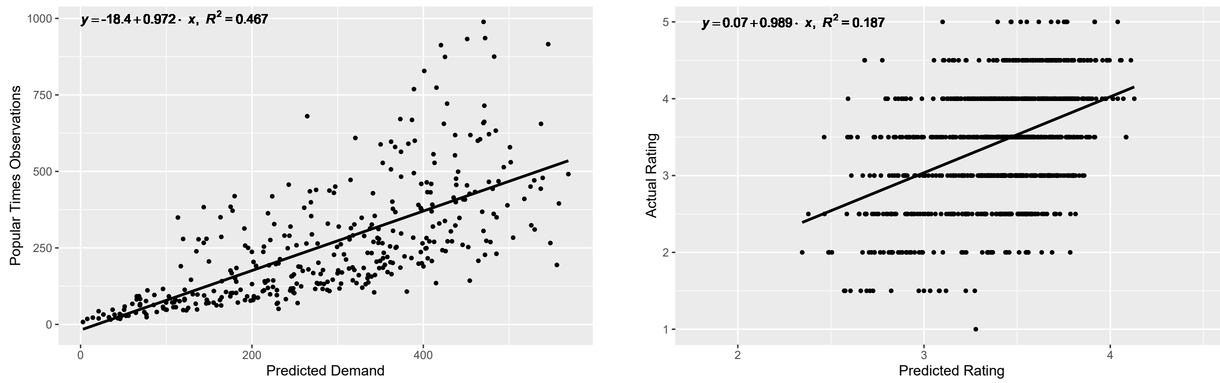


**Figure 4 Validation of demand and rating model predictions**

these two datasets to obtain a counterfactual source for bookings. (We remove fast-food chains such as McDonald's as they rarely receive reviews.)

As the Google data is by occupancy over time and our demand forecasts are for number of arrivals unconstrained by capacity, we describe how we convert them for comparison. Converting the Google occupancy to number of arrivals is easy enough, as number of arrivals times average waiting time gives the area under the occupancy curve.

To calculate capacity-constrained demand forecasts, we do the following steps:

- Take the unconstrained demand estimate for restaurant $i$, $\hat{D}_i$ and distribute it according to Google *Popular Times.* curve on an hourly basis.
- For each restaurant, obtain Average Time Spent from Google and calculate hourly capacity $C_i = \frac{\text{Total seats}}{\text{Average time spent}}$, predict hourly demand using $\min\{\hat{D}_i, C_i\}$ and sum it over the open hours to obtain a constrained demand forecast.
- Repeat the procedure for each day, take the mean to find the average *daily* constrained demand for every restaurant $i$.

As for analyzing the accuracy of ratings model, we perform a holdout sample validation by estimating the ratings on a randomly chosen training set and test the estimates on a hold-out (or validation) set.

The analysis between actual versus predicted ratings (correlation coefficient is 0.428) and Google *Popular Times* observations and predicted demands (correlation coefficient is 0.683) are given in Figure 4. Moreover, the Bland-Altman plots (Bland and Altman (1999)) in Figure 5 show graphically the agreement between the estimations and Google *Popular Times* observations. The bias between the mean difference of actual and predicted ratings is almost negligible. As for the demand estimates, the bias is rather small, and on the negative side. The Google *Popular Times* observations are 21 units less than the estimated demand on a daily average of 100 customers.

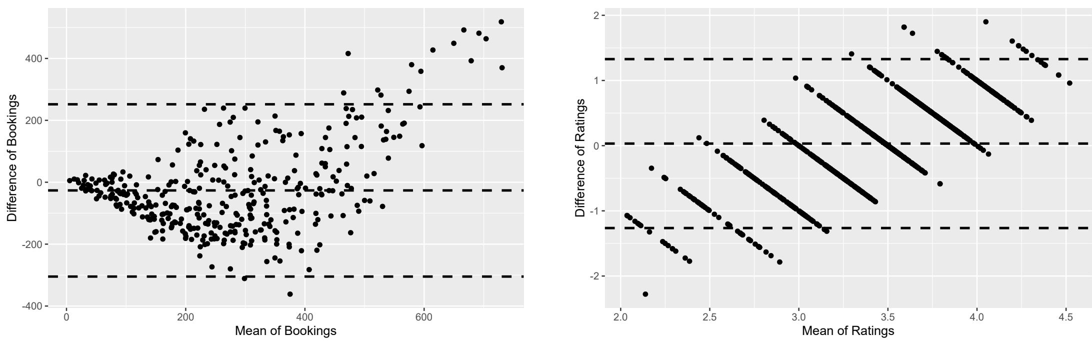


**Figure 5** **Bland-Altman plot to assess agreement between actual and predicted bookings and ratings.**

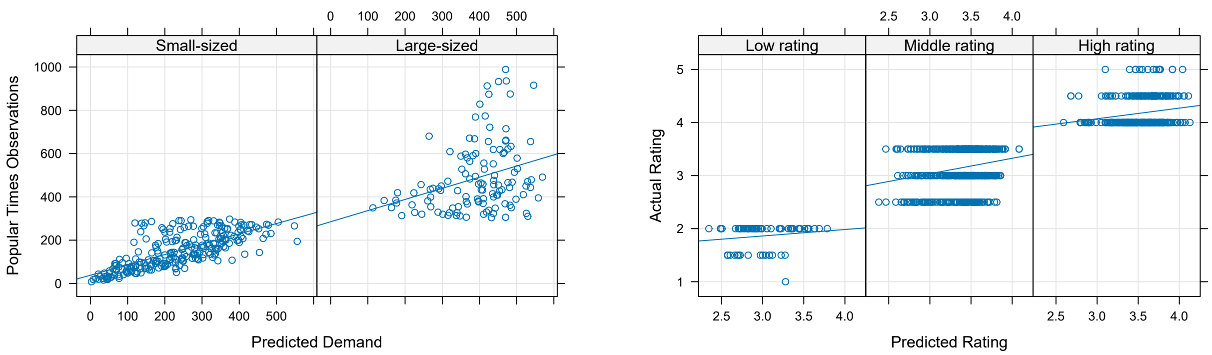


**Figure 6** **Robustness check of MNL (for demand) and LFM (for rating) applied on groups of restaurants.**

Moreover, the 95% confidence interval of the mean difference shows that most of the estimates fall within the acceptable limits.

For robustness check, we give a comparison by groupings of the restaurant. We group restaurants by stars (low, middle and high) for rating and by size (small/large; *Popular Times* observations are less/more than 300) for demand model. The results can be seen in Figure 6. We note that for small restaurants we slightly overestimate the *Popular Times* observations and for large restaurants the variance seems to be higher. The discrepancies could be for the following reasons. Recall that we use review volume as a proxy on bookings and the proportion who write reviews is not observable directly—even though we model this, it adds a layer of uncertainty. Also, observations obtained through the *Popular Times* feature does not correspond to 100% of the demand either, as it only captures users with the Android operating system and those who gave Google the permission to track.

Overall, despite severe information limitations, the demand model provides a fit of 46.7% and the rating model provides a fit of 18.7%. These are in agreement with those reported by Kabra et al. (2020) on a bike-share ridership estimation setting. Moreover, both of the prediction scenarios are robust across the categories.

## 4. Firm entry/exit model

Recall that our objective is not just to predict demand at a location for a given set of incumbents, but also to assess if a certain type of establishment will survive at a particular location, which means we need a tractable model of forward-looking competitive entry and exit.

We now describe our stylized model of entry and exit. While traditionally, especially in the economics literature, this has been formulated as a simultaneous or dynamic game, we depart from this game-theoretic modeling approach, both for tractability as well as to avoid strong assumptions on beliefs and information.

Specifically, we do not model agents and their actions. They do exist in the background, but we model only the *results* of their actions. This is done for two practical reasons: First, the number and nature of the agents (potential entrants) are, in our context at least, unknown not only to the other potential entrants but also to the researcher wanting to validate the model empirically; so this adds a layer of complexity to strategic models which are already difficult to solve. Second, even if the agents were known to all, game-theoretic modeling would require assumptions on their beliefs that, for tractability, are often drawn from a common and known distribution. However, the time-scale of retail-shop location events is years so the assumptions on common belief distributions are especially strong in our context, as they imply beliefs on future competition. As we mentioned in §1.1, our non-strategic modeling has precedents and echoed in the literature (Prescott and Visscher (1977), Pancras et al. (2012)).

Thus, in our reduced-form model agents and their actions shall remain in the background and we are concerned only with the outcomes (entry, exit or no change). We model probabilities of such outcomes as a stochastic process that is a function of a configuration of incumbents. The long-term value of a location is influenced by future competition and we model this by the probability that *some* firm will open a facility or an incumbent will quit, and the rates of entry and exit are proportional to the relative profit or loss of a location in a recursive formulation (as entry and exit rates of the other firms in the future influence these probabilities). The resulting dynamic program cannot be solved exactly, so calculating the value functions remains difficult even in our model, and we have to resort to approximation or iterative methods if we want to calculate the value functions.

The concept we are interested in, representing an eventual outcome, is of a limiting-state—a configuration of locations that remain static, where the economic motivation for an entry or an exit is null. This is somewhat distinct from equilibria in actions that is studied in economics. We next present the notation and then our model.

### 4.1. Notation

Time is discrete and in any period (say, a period of one day) *at most* one event (entry or exit) occurs. To reduce notation, we let a facility of *characteristic* $\ell$ represent a combination of location $g$, type $c$, features vector $\boldsymbol{y}$ and price-point $p$, that is, $\ell = (g, c, \boldsymbol{y}, p)$. Locating *at* $\ell$ should be understood henceforth as both the physical location as well as the design characteristics. We denote all possible characteristics as a finite set $\mathcal{N}$ (which implicitly includes the $\emptyset$, or not locating anywhere).

Costs in our model are of two different types, fixed (independent of demand) and variable (say linear in demand). The fixed costs are further classified as one-time costs, $F_\ell$, to set up the facility at $\ell$, say construction or renovation or goodwill payment cost, and an operating fixed cost, $f_\ell$, such as rent and employee salaries. We include the dependence on characteristic $\ell$, as costs are likely to depend on location and features as well as the price-category.

Our state space is a 0-1 vector $\boldsymbol{n}$ (of size $|\mathcal{N}|$) whose elements represent characteristics $\ell$, and the $\ell^{th}$ position $\boldsymbol{n}_\ell = 1$ if there exists a facility with characteristic $\ell$. If we want to specify the vector at time $t$, we denote it by $\boldsymbol{n}^t$ and call it facility *configuration* at time $t$; when it is clear from the context we drop the $t$ index to make the notation easier to read.

We use the indicator function $\mathbb{1}_{[\ell]}$ to represent a vector with a 1 in the position corresponding to $\ell$. For instance, when $\boldsymbol{n}^{t-1}$ is the current configuration at time $t-1$ and if a new facility opens with a feature set $\ell$, we update the vector as $\boldsymbol{n}^t = \boldsymbol{n}^{t-1} + \mathbb{1}_{[\ell]}$ to represent the new profile.

Note that by our choice of notation we are prohibiting the possibility of two exactly identical facilities appearing at the same location, i.e., identical in all respects. This is not such a restrictive assumption as we can expand the characteristics set trivially. A similar convention is adopted in Hotelling (1929) and Prescott and Visscher (1977).

We let $\mathcal{I}_{\boldsymbol{n}^t}$ to represent the set of incumbents, at time $t$, corresponding to all the 1's in the vector $\boldsymbol{n}^t$. The set of potential locations is the complement of this set that we represent by $\mathcal{P}_{\boldsymbol{n}^t}$. So $\mathcal{I}_{\boldsymbol{n}^t} \cup \mathcal{P}_{\boldsymbol{n}^t} = \mathcal{N}$. We let $v_\ell(\boldsymbol{n}^t)$ be the per-period profit if we locate at $\ell$ given the current location configuration of firms is $\boldsymbol{n}^t$. This is expressed as

$$v_\ell(\boldsymbol{n}^t) = \begin{cases} s_\ell \hat{D}_\ell - f_\ell, & \text{if } \ell \in \mathcal{I}_{\boldsymbol{n}^t}, \\ 0, & \text{otherwise,} \end{cases}$$

where $s_\ell$ is the contribution margin (spend minus the variable cost) and $\hat{D}_\ell = \sum_{g \in \mathcal{G}} M_g P_{\ell g}(\mathcal{I}_{\boldsymbol{n}^t})$, $\ell \in \mathcal{I}_{\boldsymbol{n}^t}$.

Let us denote the continuation values representing future profits (to be defined rigorously in §4.4) for a location $\ell$ at the beginning of period $t$ with state $\boldsymbol{n}^t$ as $V_\ell^t(\boldsymbol{n}^t)$. Note that this term does not include the fixed cost of locating but only operating fixed costs. Firms are interested in maximizing long-term profit and one objective could be to locate at $\ell$ that solves $\max_\ell E(\sum_{t=0}^{\infty} \rho^t v_\ell(\boldsymbol{n}^t))$, where

$\rho$ is a discount factor and the states $\boldsymbol{n}^t$ evolve according to a law determined by the entries and exits. Explicitly modeling and analyzing such a multi-agent simultaneous dynamic optimization problem however is difficult. Instead of modeling each potential entrant (without knowing how many there are), we can view this as a representative entrant who attaches uncertainty to each of the potential locations (what we refer to as the candidate set) and the entry location is a choice. Now if the uncertainty is modeled as Gumbel, the probability of choosing a location once again emerges by a proportionality rule—a location $\ell$ is chosen proportional to its long-term profitability, given by the value function of a dynamic program. We flesh out the notation and details of this process next.

### 4.2. Candidate set

At the start of every time-period there is a subset of incumbents who are making losses and are candidates for exit. We denote this subset by $\mathcal{I}^-_{\boldsymbol{n}^t} \subseteq \mathcal{I}_{\boldsymbol{n}^t}$. Likewise, a subset of potential locations shows a NPV exceeding a set-up cost, and are candidates for entry at the current state (configuration of incumbents); we denote this set of potential entry locations by $\mathcal{P}^+_{\boldsymbol{n}^t} \subseteq \mathcal{P}_{\boldsymbol{n}^t}$. These locations are akin to a consideration set where an entry/exit event can occur.

At $t$, a potential location $\ell \in \mathcal{P}^+_{\boldsymbol{n}^t}$ is a candidate for entry if $\frac{v_\ell(\boldsymbol{n}^t + \mathbb{1}_{[\ell]})}{1-\rho} \geq F_\ell$. Note that this is a somewhat restrictive assumption—a fully general/consistent model would define the candidate list based on $V^t_\ell(\boldsymbol{n}^t)$ but this would make calculating these value functions computationally intractable; hence we call our model a semi-myopic one. Likewise, an incumbent $\imath \in \mathcal{I}^-_{\boldsymbol{n}^t}$ is a candidate to exit if $v_\imath(\boldsymbol{n}^t) < 0$ (we ignore scrap costs that are modeled in Pakes et al. (2007); these can be incorporated trivially). The candidate set of entry-exit locations are the basis for our definition of a steady state $\boldsymbol{n}^\diamond$ in §5—essentially if $\mathcal{I}^-_{\boldsymbol{n}^\diamond} \cup \mathcal{P}^+_{\boldsymbol{n}^\diamond} = \emptyset$.

### 4.3. Entry/exit probabilities

In our semi-myopic model, the candidate set is defined based on profits at the current state but the entry and exit probabilities are proportional to the expected long-term profits given the current state. In short, the probability of an entry is given by:

$$p^t_\ell(V^t, \boldsymbol{n}^t) = \frac{e^{V^t_\ell(\boldsymbol{n}^t + \mathbb{1}_{[\ell]}) - F_\ell}}{A_{\boldsymbol{n}^t} + \underbrace{\sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^t}} e^{V^t_{\ell'}(\boldsymbol{n}^t + \mathbb{1}_{[\ell']}) - F_{\ell'}}}_{\text{entry}} + \underbrace{\sum_{\imath' \in \mathcal{I}^-_{\boldsymbol{n}^t}} e^{-V^t_{\imath'}(\boldsymbol{n})}}_{\text{exit}}}, \tag{5}$$

for $\ell \in \mathcal{P}^+_{\boldsymbol{n}^t}$ (0 for all $\ell \in \mathcal{P}_{\boldsymbol{n}^t} \setminus \mathcal{P}^+_{\boldsymbol{n}^t}$). And, for an incumbent, the probability of an exit is given by:

$$q^t_\imath(V^t, \boldsymbol{n}^t) = \frac{e^{-V^t_\imath(\boldsymbol{n}^t)}}{A_{\boldsymbol{n}^t} + \underbrace{\sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^t}} e^{V^t_{\ell'}(\boldsymbol{n} + \mathbb{1}_{[\ell']}) - F_{\ell'}}}_{\text{entry}} + \underbrace{\sum_{\imath' \in \mathcal{I}^-_{\boldsymbol{n}^t}} e^{-V^t_{\imath'}(\boldsymbol{n})}}_{\text{exit}}}, \tag{6}$$

for $\imath \in \mathcal{I}^-_{\boldsymbol{n}^t}$ (0 for all $\imath \in \mathcal{I}_{\boldsymbol{n}^t} \setminus \mathcal{I}^-_{\boldsymbol{n}^t}$). The term $A_{\boldsymbol{n}^t}$ governs the rate of entry or exit and can be calibrated based on observed activity in the market. It depends only on $\boldsymbol{n}^t$ (i.e., the state, or the current incumbents, but not their value functions). One would expect that as the incumbent set gets larger, the number of attractive locations become fewer. This is inherently captured by the values of the potential locations decreasing, but this constant allows better control of the rate of openings. A few possibilities are $A_{\boldsymbol{n}^t} = A$, a constant; $A_{\boldsymbol{n}^t} = |\boldsymbol{n}^t|A$, depending only on the number of incumbents; $A_{\boldsymbol{n}^t} = A + e^{\sum_{\ell \in \mathcal{I}_{\boldsymbol{n}^t}} v_\ell(\boldsymbol{n}^t)}$, a more elaborate incorporation of the incumbents' (current period) profits influencing entry. It can potentially change over time also, but we leave it as a stationary function for tractability.

To justify this modeling choice, for one, it makes perfect sense that the firm making bigger losses has a higher probability of exiting compared to the one making smaller losses. Likewise the potential profits at a location should dictate the chances that *some* firm locates there. As we mentioned earlier, we do not model each individual firm's decision process, but only the outcome configuration. By using such reduced-form model of entry and exit probabilities, we side-step a vexing problem in modeling entry decisions—we do not need any assumption on the number or nature of firms contemplating entry or their (common) beliefs on transition probabilities. The latter is generally agreed to be a far-fetched assumption, but is considered unavoidable in low-level strategic game-theory models.

Remark 1. We can interpret the above probabilities as follows: There are random factors that influence the costs at each location, and these random variables are governed by a Gumbel distribution. Namely (suppressing the time index $t$):

$$\max\{\underbrace{\ln A_{\boldsymbol{n}^t} + \xi_0}_{\text{no event}}, \underbrace{\max_{\ell \in \mathcal{P}^+} V_\ell(\boldsymbol{n}^t) - F_\ell + \xi_\ell}_{\text{entry event}}, \underbrace{-\max_{\imath \in \mathcal{I}^-} V_\imath(\boldsymbol{n}^t) + \xi_\imath}_{\text{exit event}}\}.$$

In this addition of random variables to model entry or exit at a location, we coincide with the modeling in Aguirregabiria and Vicentini (2016) (see for instance, their Equation (23)). However, there is an important difference. While the random term in Aguirregabiria and Vicentini (2016) (conf. their Equation (14)) is firm and action-specific, our $\xi$ is specific to each of the locations and common to the firms and drawn independently in each period.

We observe the outcome of the maximization event after the realization of the unobserved random variables. Thus we are able to reconcile both the maximization of long-term profits by the firms as well as the fact that one observes at most one entry or exit in a small time period.

### 4.4. The value functions

The states evolve according to a categorical process over the candidate set of locations (and the null event of no-change) with probabilities given in (5,6). The value functions for each location are then given by a recursive equation.

The quantity $V_\ell^t(\boldsymbol{n}^t)$ represents the discounted future profit for each location $\ell$, whether there is a facility at that location or it is vacant, and influences the probability of a facility locating there (if there is none at present), or exiting from the location (if already located). If $\ell \notin \mathcal{I}_{\boldsymbol{n}^t}$, its value in the current period $v_\ell(\boldsymbol{n}^t) = 0$, but it could still have $V_\ell^t(\boldsymbol{n}^t) > 0$ as this captures value from a future opening.

The value functions are calculated according to the following set of equations:

$$\begin{aligned} V_\ell^t(\boldsymbol{n}^t) = v_\ell(\boldsymbol{n}^t) + \rho \Bigg[ & \underbrace{\left(1 - \sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^{t+1}}} p_{\ell'}^{t+1}(V^{t+1}, \boldsymbol{n}) - \sum_{\imath \in \mathcal{I}^-_{\boldsymbol{n}^{t+1}}} q_\imath^{t+1}(V^{t+1}, \boldsymbol{n}^t)\right) V_\ell^{t+1}(\boldsymbol{n}^t)}_{\text{No change in state}} \\ & + \underbrace{\sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^{t+1}}} p_{\ell'}^{t+1}(V^{t+1}, \boldsymbol{n}^t) V_\ell^{t+1}(\boldsymbol{n}^t + \mathbb{1}_{[\ell']})}_{\text{entry}} + \underbrace{\sum_{\imath \in \mathcal{I}^-_{\boldsymbol{n}^{t+1}}} q_\imath^{t+1}(V^{t+1}, \boldsymbol{n}) V_\ell^{t+1}(\boldsymbol{n} - \mathbb{1}_{[\imath]})}_{\text{exit}} \Bigg]. \end{aligned} \tag{7}$$

Note that the $V$'s form a matrix with rows corresponding to locations and the columns to all possible states. And that there is no explicit optimization—they are essentially expected value calculations based on the profits obtained in each period.

*If* a Bellman equation were to exist, it would be a recursion for stationary value functions as follows $\forall \ell \in \mathcal{I}_{\boldsymbol{n}^t}$ (define $V_\ell(\boldsymbol{n}^t) = 0, \ell \notin \mathcal{I}_{\boldsymbol{n}^t}$):

$$V_\ell(\boldsymbol{n}^t) = v_\ell(\boldsymbol{n}^t) + \rho \left[ \underbrace{\left(1 - \sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^t}} p_{\ell'}(V, \boldsymbol{n}^t) - \sum_{\imath \in \mathcal{I}^-_{\boldsymbol{n}^t}} q_\imath(V, \boldsymbol{n}^t)\right) V_\ell(\boldsymbol{n}^t)}_{\text{No change in state}} + \underbrace{\sum_{\ell' \in \mathcal{P}^+_{\boldsymbol{n}^t}} p_{\ell'}(V, \boldsymbol{n}^t) V_\ell(\boldsymbol{n}^t + \mathbb{1}_{[\ell']})}_{\text{entry}} + \underbrace{\sum_{\imath \in \mathcal{I}^-_{\boldsymbol{n}^t}} q_\imath(V, \boldsymbol{n}^t) V_\ell(\boldsymbol{n}^t - \mathbb{1}_{[\imath]})}_{\text{exit}} \right], \tag{8}$$

with the entry and exit probabilities defined as in (5,6), and hence themselves functions of $V$. The sets $\mathcal{P}_{\boldsymbol{n}}, \mathcal{P}^+_{\boldsymbol{n}}, \mathcal{I}_{\boldsymbol{n}}, \mathcal{I}^-_{\boldsymbol{n}}$ as mentioned in §4.2 do not depend on our value functions but only on the state $\boldsymbol{n}^t$. The question then would be is there a solution $V_\ell(\boldsymbol{n})$ to these set of equations?

We do not know the answer in the general case (where the candidate set for entry/exit as well as the transition probabilities depend on the value functions), but we can show the stationary value functions exist under some (restrictive) assumptions.

ASSUMPTION 1. *Let* $M_\ell = \sum_{g \in \mathcal{G}_\ell} M_g$, *and a weighted mean distance for* $\ell$ *as* $\bar{d}_\ell = \sum_{g \in \mathcal{G}_\ell} \frac{M_g}{M_\ell} d_{\ell,g}$, *where* $d_{\ell,g}$ *is the distance from* $g$*'s center to* $\ell$*. We define the attractiveness of a facility by the average utility of a customer's visiting* $\ell$*, given by* $W_\ell = \beta_d \bar{d}_\ell + \beta_r r_\ell + \beta_p p_\ell$.

Let $P_\ell(\mathcal{I})$ be the probability the customer visits $\ell$, as per the MNL probability law defined by Equation (1). The demand the firm expects is then given by $\hat{D}_\ell = M_\ell P_\ell(\mathcal{I})$.

PROPOSITION 1. *Under Assumption 1 and the operating fixed costs $f_\ell = 0$ (hence no exits), the stationary value functions $V$ exist for the system (8).*

Among all the states, based on the stationary value functions, some states $\boldsymbol{n}^\diamond$ are special and define our steady state: They are the ones that satisfy $\mathcal{I}^-_{\boldsymbol{n}^\diamond} \cup \mathcal{P}^+_{\boldsymbol{n}^\diamond} = \emptyset$. Such a configuration of facility locations $\boldsymbol{n}^\diamond$, once reached, would not see a change as the entry and exit probabilities are all 0. Calculating the value functions exactly is challenging because of the state-space explosion, butfuture research could explore approximation methods.

## 5. Location equilibria

In a competitive setting, predicting whether a system evolves to an equilibrium, and if so which one, is complicated. Indeed, various notions of equilibrium exist in the literature, often developed for a specific context keeping analytical and computational tractability in mind. Their predictive power and insight depend on whether they can be easily characterized, whether they are unique, whether they always exist, and so on. Information, strategic decisions of the players as well as stochastic elements in the model complicate both analysis and computations.

In this part, we define an equilibrium steady-state concept for location selection. It abstracts away the dynamic strategies of the firms without the need to know the number of firms contemplating entry. We analyze its properties and give a computational procedure using IP for finding it.

### 5.1. Location steady state and analysis

While the states may evolve stochastically, our steady-state is deterministic and refers to a configuration of locations rather than the actions of players (as would say in a strategic game). We define a set of incumbent locations $\mathcal{I}^\diamond$ to be in steady state when none of the incumbents has an incentive to exit (i.e., are making positive profits) and potential firms have no incentive to open a new retail facility at any of the candidate locations as the long-term profit from such a venture, given $\mathcal{I}^\diamond$, is less than the fixed opening costs; in other words, the candidate set is empty. So a *steady-state* is a set of facilities $\mathcal{I}^\diamond = \mathcal{I}_{\boldsymbol{n}^\diamond}$ such that $\frac{v_\ell(\boldsymbol{n}^\diamond + \mathbb{1}_{[\ell]})}{1-\rho} \leq F_\ell$, $\forall \ell \in \mathcal{P}_{\boldsymbol{n}^\diamond}$ and $\forall \imath \in \mathcal{I}_{\boldsymbol{n}^\diamond}$, $v_\imath(\boldsymbol{n}^\diamond) \geq 0$. Note that in this steady-state there are no incentives for entry or exit, so the value functions coincide with the simple net-present values of revenues.

Why is this of interest to a firm? If the firm is considering opening at a location, it would want to survive for a long time. One way of formalizing this is that it is open in steady state; the number and nature of which, however, can change depending on where it locates. Therefore predicting equilibria if the firm enters at a location is still a challenging problem, we show that it can be formulated as an integer linear program in §5.2 (as opposed to calculating solutions of systems of dynamic programs).

We next show that under a slightly simplified model of demand a steady state always exists. For simplicity and to conform to mathematical programming conventions, we use $i, l$ for incumbent and potential locations respectively rather than the special symbols $\imath$ and $\ell$ we have been using thus far.

Consider a set of grid points $\mathcal{G}_l$ as a catchment area for a potential location $l$ under consideration, say all the grid points within a certain distance from $l$. We specify now a simplified model of demand, that allows us to linearize constraints in an IP to compute a steady state. Let $M_l$ represent the income-adjusted population density covering the catchment area of $l$, as per Assumption 1.

We let $P_l(\mathcal{I}^\diamond)$ be the probability the customer visits $l$, as per the MNL probability law defined by Equation (1). The demand the firm expects is then given by $\hat{D}_l = M_l P_l(\mathcal{I}^\diamond)$.

Now, once in steady state $\mathcal{I}^\diamond$, since no entry or exits occur, the continuation value is easy to calculate. Since the per-period profit is $v_l(\mathcal{I}^\diamond) = s_l M_l P_l(\mathcal{I}^\diamond) - f_l$, then the infinite-horizon profit for location $l \in \mathcal{I}^\diamond$ is

$$V_l(\mathcal{I}^\diamond) = v_l(\mathcal{I}^\diamond)/(1-\rho).$$

For a $\boldsymbol{n}$ to be in steady state $\mathcal{P}^+_{\boldsymbol{n}} = \emptyset$ and $\mathcal{I}^-_{\boldsymbol{n}} = \emptyset$, and hence this can be specified by the following two inequalities respectively:

$$\text{No entries:} \qquad \frac{1}{1-\rho}\left[\frac{s_l M_l e^{W_l}}{1+e^{W_l}+\sum_{j\in\mathcal{I}^\diamond} e^{W_j}} - f_l\right] \le F_l \qquad \forall l \notin \mathcal{I}^\diamond, \tag{9}$$

and

$$\text{No exits:} \qquad \frac{s_i M_i e^{W_i}}{1+\sum_{j\in\mathcal{I}^\diamond} e^{W_j}} \ge f_i \qquad \forall i \in \mathcal{I}^\diamond. \tag{10}$$

These conditions are best viewed in a slightly different way. Let $\tilde{w}_i = e^{W_i}$ for all $i \in \mathcal{I}^\diamond$. Then,

$$\left[\frac{(s_l M_l - f_l - (1-\rho)F_l)\tilde{w}_l}{f_l + (1-\rho)F_l}\right] \le 1 + \sum_{j\in\mathcal{I}^\diamond} \tilde{w}_j \qquad \forall l \notin \mathcal{I}^\diamond, \tag{11}$$

and

$$\frac{s_i M_i \tilde{w}_i}{f_i} \ge 1 + \sum_{j\in\mathcal{I}^\diamond} \tilde{w}_j \qquad \forall i \in \mathcal{I}^\diamond. \tag{12}$$

For all $i \in \mathcal{I}^\diamond$, let

$$\tilde{a}_i = \frac{(s_i M_i - f_i - (1-\rho)F_i)\tilde{w}_i}{f_i + (1-\rho)F_i}, \qquad \text{and} \qquad \tilde{b}_i = \frac{s_i M_i \tilde{w}_i}{f_i}.$$

Also, note that $\tilde{a}_i \le \tilde{b}_i$. For any set $\mathcal{I}$, let $\tilde{w}_{\mathcal{I}} = \sum_{j\in\mathcal{I}} \tilde{w}_j$.

PROPOSITION 2. *A steady state, i.e., a set of locations where there is no incentive to exit or enter, always exists.*

Corollary 1. *There exists at least one path (a sequence of entries and exits with positive probabilities) from any set of incumbents to a steady state.*

Define a $\tilde{a}_{\max}$ policy as follows: For any incumbent set, locate at a feasible location with the maximum value of $\tilde{a}$; if there is no feasible location, wait till such a location appears. As a corollary, the proof of Proposition 2 gives the following guidance for firms:

Corollary 2. *Given any set of incumbent locations, a firm wishing to maximize survival should locate according to the $\tilde{a}_{\max}$ policy. In other words, if the firm locates at $i$ according to the $\tilde{a}_{\max}$ policy, then any steady state that results from the current set of incumbents will contain $i$.*

### 5.2. A firm's decision problem for survivability

How can a firm contemplating entry make a decision if it wants to survive in the long-term? This subsection takes a practical point of view of such a firm, which wants to find a safe profitable long-term location—and is not concerned about how such a steady-state arises.

The firm's objective is to locate at a "safe" location—which may not necessarily be the current most profitable location—defined as a location that would exist (once it enters) in steady state. The difficulty, of course, is that there could be many equilibria and finding them is computationally quite challenging. The decision of the firm to enter at a location may itself influence what steady state emerges further complicating the calculations.

Researchers have proposed a range of computational methods for finding a steady state facility location configuration: solving a system of nonlinear equations by numerical methods, or using Gaussian approximations, other stochastic approximations, as well as homotopy methods; see Doraszelski and Pakes (2007) for an overview. In general, all of them require complicated implementations and tuning to achieve convergence, even to find an approximate equilibrium.

In contrast, partly because of our simplified steady state definition, we can resort to an IP formulation to calculate the equilibria and determine a long-run optima (that we define as the "safest" amongst all the possible equilibria) for an entrant. The advantages of IP as a solution method cannot be underestimated, as a firm can use powerful off-the-shelf solvers to gain insight into location decisions.

Let variables $x_j = 1$ if there is a facility at $j$ in steady-state and 0 otherwise. Feasible solutions to the IP correspond to a steady state. Our focus is on finding an optimal location from the firm's point-of-view with an objective of long-term survival. The fact that the firm decided to locate at a certain location influences the eventual equilibria, so the location decision has to take this into consideration. Our premise is that the firm would want to locate to maximize the chances that it will lead to an equilibria which contains that location (the "safe" location).

To capture this, let the 0-1 variable $\tilde{x}_i$ represent the firm's *decision* to locate at i. The binary variable $x_i$ in contrast represents whether the location $i$ survives in the long-term. Whether there is such a steady state and what configuration will emerge, of course, depends on the fact that $\tilde{x}_i = 1$. The firm would want to only locate so that if $\tilde{x}_i = 1$ then in steady state if $x_i = 1$; an even safer objective is a max-min objective as we explain below where it also wants the location $i$ to be as "safe" as possible.

First, we devise constraints that represent equilibria. The first set of constraints are linearizations which ensure that a location $l$ *without* a facility ($x_l = 0$) is not attractive enough for anyone to enter; i.e., should satisfy (the summation in the denominator is over all locations):

$$\frac{1}{1-\rho}\left[\frac{s_l M_l (1-x_l)\tilde{w}_l}{1+(1-x_l)\tilde{w}_l + \sum_{j\neq l} x_j \tilde{w}_j} - f_l\right] \le F_l \qquad \forall l \notin \mathcal{I}^{\diamond}. \tag{13}$$

Note that the $\tilde{w}_i$'s can be treated as constants as they are the customer's utility functions based on distance, ratings etc. and depend only on $i$, and this constraint is trivially satisfied when $x_l = 1$.

Similarly, a location $i$ with a facility $x_i = 1$ should be such that there is no incentive to exit, or

$$\frac{s_i M_i x_i \tilde{w}_i}{1+\sum_j x_j \tilde{w}_j} \ge x_i f_i \qquad \forall i \in \mathcal{I}^{\diamond}. \tag{14}$$

We now convert both of them to a linear IP. Equation (13) can be linearized by simple algebra into constraint (17) and Equation (14) below can be linearized by the big-$M$ method into (18).

The firm first wants to locate which survives in steady state, but there could be multiple equilibria once it locates—a conservative objective is that it survives under all equilibria—the minimum gap between the NPV of its operating profits and its fixed investment cost among all possible equilibria is positive. To find a single location, subject to steady state constraints, the firm should find a location $\imath \in \mathcal{P}$ such that the below is positive.

$$\frac{1}{1-\rho}\left(\frac{s_\imath M_\imath \tilde{w}_\imath}{1+\sum_j x_j \tilde{w}_j} - f_\imath\right) - F_\imath = \frac{1}{1-\rho}\left(\frac{s_\imath M_\imath \tilde{w}_\imath}{1+\sum_j x_j \tilde{w}_j} - [f_\imath + (1-\rho)F_\imath]\right). \tag{15}$$

The problem can be solved by checking if the following integer program is feasible for each fixed location $\imath$, $\imath \in \mathcal{P}$:

$$\min_{Z_\imath} \quad Z_\imath \tag{16}$$

$$\text{s.t.} \quad (1-x_i)s_i M_i \tilde{w}_i - f_i\left(1+(1-x_i)\tilde{w}_i + \sum_{j\neq i} x_j \tilde{w}_j\right) \le F_i(1-\rho)\left(1+(1-x_i)\tilde{w}_i + \sum_{j\neq i} x_j \tilde{w}_j\right), \quad \forall i \tag{17}$$

$$(1+\sum_j \tilde{w}_j)(1-x_i) + x_i s_i \frac{M_i}{f_i}\tilde{w}_i \ge 1 + \sum_j x_j \tilde{w}_j, \quad \forall i \tag{18}$$

$$\tilde{x}_\imath s_\imath M_\imath \tilde{w}_\imath - (1+\sum_j x_j \tilde{w}_j)(f_i + (1-\rho)F_\imath) \le Z_\imath, \tag{19}$$

$$1 = \tilde{x}_\imath \le x_\imath, \quad x_j \in \{0,1\}. \tag{20}$$

For a particular $\imath$ if the above system has an optimal solution with value $Z_\imath^* > 0$, then we can conclude that *every* steady-state solution arising from locating there would lead to a profitable solution—in other words, $\imath$ is a "safe" location. The firm can run this math program for all the locations it is considering and pick the best "safe" location if it has other criteria in addition. Extending this to finding $k$ safe locations is challenging because of the combinatorial explosion. We leave it as an interesting future research direction.

### 5.3. Entry/exit model output example

Once we estimate ratings and demand model parameters from data, we can use them in our IP model for the decision problem. Running the IP is fairly routine and we show a representative set of results in this section.

We focus on two specific zip code regions in our example city, Las Vegas, to illustrate the nature of the IP results and choose Mexican cuisine in two price categories as an example. Although our entry/exit model is general, we restrict to one cuisine and features in our exercise to be able to visualize the actions clearly in the figures. Figure 7 (top-left) shows the existing restaurants with a triangle sign obtained from Yelp. Income-adjusted population density (market size) is color coded around the grid points. The numbers in the grid points are the predictions of utility for each individual restaurant $\tilde{w}_j$.

In this specific example, one can clearly observe that the zip-code on the west side is higher in market size compared to the east side. Moreover, the west region is predicted to favor Mexican cuisine more than its east side neighbor as indicated by the attraction values. Figure 7 provides the IP solution for current decision to locate $\tilde{x}_i$ (top-right) (in order to gain insight into solution, we plot top 5 locations) and for survival state locations $x_i$ (bottom). According to the steady state solution given by the IP, some of the restaurants would shut down and a few of the restaurants tend to open at locations where the attractiveness of the restaurant is predicted to be higher with a higher market size in its catchment area. Also, two incumbent restaurants would survive while two shut down. As in Prescott and Visscher (1977), our optimization result indicates that firms tend to space (or differentiate) themselves rather than cluster as in the Hotelling model—so the conclusions of the latter strongly depend on the moveability of the facilities.

## 6. Conclusion

We study the problem of service facility location when firms anticipate future entries and exits. Our goal is to operationalize competitive service facility-location models based on customer choice. Traditional market-research based on focus groups or surveys is difficult, time-consuming, expensive and not very reliable as (a) finding representative samples is not easy, (b) a future service product is difficult to visualize. Therefore, we believe data-based modeling is the most promising alternative.

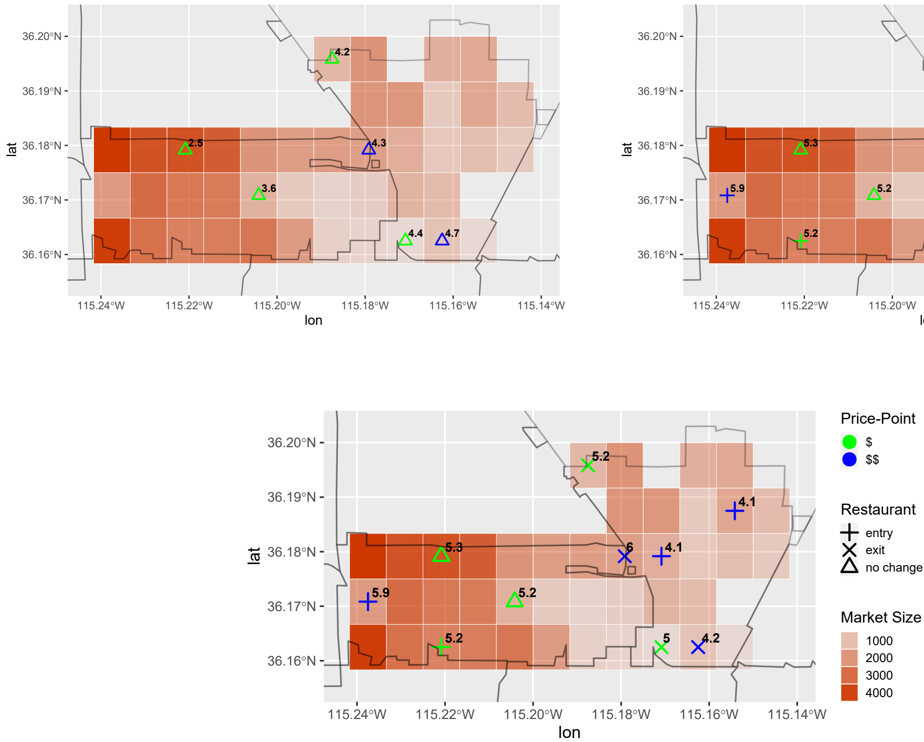


**Figure 7** **Incumbents, current decision to locate, $\tilde{x}_i$ and survival state locations $x_i$.**

We show that by crossing public datasets on density, demographics, review ratings and with some minor calibration of a single parameter from a sample of scraped data, we can very accurately predict ratings as well as demand. We demonstrate our findings for the restaurant industry where estimation is particularly challenging because demand is a function of non-quantifiable measures such as quality and customer tastes.

Our study provides an easy-to-implement computational tool which can be immensely useful for location analysis in retail industries. We show, via numerical analysis, *improved* location can increase chances of survival significantly. Also, we gain insight into the trade-off between review ratings and location-centrality of a restaurant—we find the first factor has a higher impact on demand.

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# E-Companion for "Popularity Prediction and Location Analytics Based on Public Data: Application to Restaurants"

This e-companion is intended to supplement the paper. It includes the proofs along with additional material on the model.

## EC.1. Proofs

In this section, we present the proofs to the steady state model.

### EC.1.1. Proof of Proposition 1

We first note that (8) can be written in terms of independent Type-I Extreme-Value random variables $\xi_0, \{\xi_{\ell'}\}_{\ell'\in\mathcal{P}_{\boldsymbol{n}}}, \{\xi_\imath\}_{\imath\in\mathcal{I}_{\boldsymbol{n}}}$ as follows:

$$V_\ell(\boldsymbol{n}) = v_\ell(\boldsymbol{n}) + \rho \mathsf{E}_{X^V}[V_\ell(\boldsymbol{n} + X^V(\boldsymbol{n}))]. \tag{EC.1}$$

The random variable $X^V(\boldsymbol{n})$ is a random indicator vector of size $|\mathcal{N}|$ (including an all-zero vector indicating $\emptyset$) giving the opening or closing of at most one location. It depends on the state $\boldsymbol{n}$ and the value functions $V$ as follows:

$$X^V(\boldsymbol{n}) = \begin{cases} \mathbb{1}_{[\ell']} & \ell'\in\mathcal{P}_{\boldsymbol{n}}, \text{ with Prob } \{\ell' = \arg\max\{\ln A + \xi_0, \{V_{\ell'}(\boldsymbol{n}+\mathbb{1}_{[\ell']}) - F_{\ell'} + \xi_{\ell'}\}_{\ell'\in\mathcal{P}^+_{\boldsymbol{n}}}, \{-V_\imath(\boldsymbol{n}) + \xi_\imath\}_{\imath\in\mathcal{I}^-_{\boldsymbol{n}}}\} \\ -\mathbb{1}_{[\imath]} & \imath\in\mathcal{I}_{\boldsymbol{n}}, \text{ with Prob } \{\imath = \arg\max\{\ln A + \xi_0, \{V_{\ell'}(\boldsymbol{n}+\mathbb{1}_{[\ell']}) - F_{\ell'} + \xi_{\ell'}\}_{\ell'\in\mathcal{P}^+_{\boldsymbol{n}}}, \{-V_\imath(\boldsymbol{n}) + \xi_\imath\}_{\imath\in\mathcal{I}^-_{\boldsymbol{n}}}\} \\ \mathbf{0} & \text{with Prob } \{0 = \arg\max\{\ln A + \xi_0, \{V_{\ell'}(\boldsymbol{n}+\mathbb{1}_{[\ell']}) - F_{\ell'} + \xi_{\ell'}\}_{\ell'\in\mathcal{P}^+_{\boldsymbol{n}}}, \{-V_\imath(\boldsymbol{n}) + \xi_\imath\}_{\imath\in\mathcal{I}^-_{\boldsymbol{n}}}\} \end{cases}$$

Here, given there are $|\mathcal{N}|$ locations, $V: 2^{|\mathcal{N}|} \to \mathbb{R}^{|\mathcal{N}|}$ can be considered a matrix. The superscript $V$ in $X^V$ reminds us that the random variable is a function of $V$ (and $\boldsymbol{n}$). We also write $m \in \boldsymbol{n}$ to indicate that a location $m$ is open in state $\boldsymbol{n}$.

The standard modus operandi for proving the existence of the Bellman equation does not work however as the operator on the right-hand side of (EC.1) depends endogenously on the value function and is not a contraction operator (as far as we know). However, for the case of no-exits, under our entry probability model of (5) it is easy to show the existence of stationary value functions.

Let $\mathcal{L}_j$ represent all configurations with $j$ open facilities.

Consider for $\boldsymbol{n} \in \mathcal{L}_N$. Trivially, since all locations have a facility, and there are no exits, $V_\ell(\boldsymbol{n}) = \frac{\tilde{w}_\ell}{(1-\rho)\tilde{w}(\boldsymbol{n})}$. Assume stationary value functions exist for all configurations of size $j+1$. The entry probabilities (without exits) are given by

$$p^t_\ell(V^t, \boldsymbol{n}) = \frac{e^{V^t_\ell(\boldsymbol{n}+\mathbb{1}_{[\ell]}) - F_\ell}}{A_{\boldsymbol{n}} + \sum_{\ell'\in\mathcal{P}^+_{\boldsymbol{n}^t}} e^{V^t_{\ell'}(\boldsymbol{n}+\mathbb{1}_{[\ell']}) - F_{\ell'}}}, \tag{EC.2}$$

Note that all the configurations on the right-hand side have size one greater than that of $\boldsymbol{n}$.

Then, for a $\boldsymbol{n} \in \mathcal{L}_j$, the operator (EC.1)'s transition probability law is governed by the stationary value functions of configurations in $\mathcal{L}_{j+1}$ and $A_{\boldsymbol{n}}$ (which does not depend on the value functions) only. Hence a stationary solution to (EC.1) is given by a recursive equation. Q.E.D.

### EC.1.2. Proof of Proposition 2

We can gain insight into the equilibria because of the simplicity of our model. Figure EC.1 gives us intuition into the nature of the equilibria.

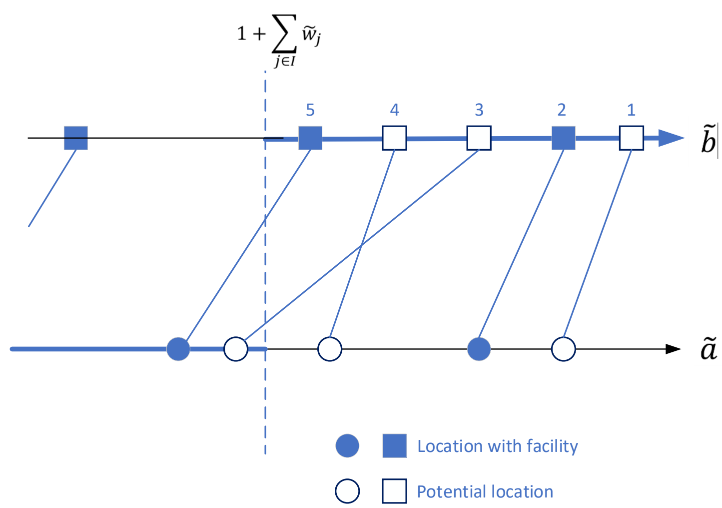


**Figure EC.1** **Locations (squares) can survive only if their $\tilde{b}_i$ is to the right of $\tilde{w}_{\mathcal{I}}$ on the $\tilde{b}$ dimension; New entrants enter only if the corresponding $\tilde{a}_i$ value (circle) is to the right of $\tilde{w}_{\mathcal{I}}$ on the $\tilde{a}$ dimension. So a steady state is simply where every black pair of nodes will be to the right of $\tilde{w}_{\mathcal{I}}$ and all pairs to the left would be white nodes.**

We prove by constructing a steady state from the null set (no incumbent facilities). Order all the locations according to the values of their $\tilde{b}_i$ as in Figure EC.1. Their corresponding $\tilde{a}_i$ values are on the bottom $\tilde{a}$ line, and the circle and squares joined by a line representing each location $i$. Let $\mathcal{I}^t$ be the set of open locations in step $t$ with $\mathcal{I}^0 = \emptyset$. Consider the following algorithm. At round $t+1$, locate at the right-most open location $i$ (largest $\tilde{b}_i$) on the $\tilde{b}$ dimension, such that $\tilde{a}_i \geq 1 + \tilde{w}_{\mathcal{I}^t}$. Note that this might mean that we do not necessarily locate sequentially from right to left on the $\tilde{b}$ axis (for instance node 3) in the Figure EC.1. If such an $i$ exists, let $\mathcal{I}^{t+1} = \mathcal{I}^t \cup \{i\}$. If not, we claim we have a steady state.

Let $t^\diamond$ be the terminating round (clearly this process is finite). To show that it terminates at a steady state, we have to show that steady state conditions hold. We first show that $\tilde{w}_{\mathcal{I}^{t^\diamond}}$ is monotonic increasing and no incumbent at $t^\diamond$ exits.

Suppose $i_{t+1}$ is added in round $t+1$. By definition, for all $i \in \mathcal{I}^{t+1}$, $\tilde{b}_{i_{t+1}} - \tilde{w}_{i_{t+1}} \geq \tilde{a}_{i_{t+1}} \geq 1 + \tilde{w}_{\mathcal{I}^t}$. Or, $\tilde{b}_{i_{t+1}} \geq \tilde{w}_{\mathcal{I}^{t+1}}$. Since all $i \in \mathcal{I}^t$ have $\tilde{b}_{i_t} \geq \tilde{b}_{i_{t+1}}$, no open location exits. So at every stage, (12) is satisfied.

Next, to prove a steady state, since we are not able to add any more $i$, we have for all $i \notin \mathcal{I}^{t^\diamond}$ and with $\tilde{b}_i \geq \tilde{w}_{\mathcal{I}^{t^\diamond}}$, we have $\tilde{a}_i < \tilde{w}_{\mathcal{I}^{t^\diamond}}$.

For $i$ such that $\tilde{b}_i < \tilde{w}_{\mathcal{I}^{t^\diamond}}$, as $\tilde{a}_i < \tilde{b}_i$, we have that $i \notin \mathcal{I}^{t^\diamond}$ we have that $\tilde{a}_i < \tilde{w}_{\mathcal{I}^{t^\diamond}}$, so the first condition for steady state (11) is satisfied. Q.E.D.

### EC.1.3. Proof of Corollary 1

Given a set $\mathcal{I}$ of incumbents, follow the state-evolution path of first closing all the locations to the left of $\tilde{w}_{\mathcal{I}}$ sequentially starting from the left. Then we follow the same strategy as in the proof of Proposition 2 starting from right-to-left along the $\tilde{b}$ dimension, and closing any location that falls to the left of $\tilde{w}_{\mathcal{I}}$ at every stage. Same arguments show that this ends in a steady state. Q.E.D.

## EC.2. DP methods to calculate value functions

The best strategy to locate can be found based on DP methods. However, the recursions are impossible to solve for any but trivial state-evolutionary laws as the state-space explodes, even for the single-location case (recall that facility type and price categories are also our design variables). In the following we illustrate this challenging problem through naive examples. First, we explain how we decide on the parameters of the model.

### EC.2.1. The rates of entry/exit and discount factor

We estimate the rate of entry and exit from the Yelp data. We illustrate this procedure with an example. Say, we want to find the best location for a specific cuisine type (e.g. American) in a certain zip code (e.g. 89102). In this example, we also assume the business owner has an investment plan to open a restaurant of price point \$ and 5000 sq-ft size.

We consider the reviews for all the restaurants of this specific type in Yelp dataset. A column named "open" shows whether a specific restaurant currently is operating or not (Table EC.1). Based on this information, we separate the restaurants in two groups: open and closed. Among the open/closed ones, we select the first/last review left for the restaurant and note the corresponding review date. This serves as an approximation for the specific restaurant opening/closing time (Tables EC.2, EC.3). Later, we sort the restaurants according to the first/last review they received. Then, simply summing the time intervals in the open/closed set for all the restaurants, we find the rate of entry/exit. Based on this, we can decide on the parameter $\boldsymbol{A_n}$. We use daily compounding with an interest rate of i. So, the discount factor is given by $\rho = \frac{1}{1+\frac{i}{365}}$.

**Table EC.1 A sample of restaurants in Yelp.**

| business id | user id | date | stars | open |
|---|---|---|---|---|
| 1 | 8 | 2014-10-20 | 4 | True |
| 1 | 70 | 2014-11-07 | 2 | True |
| 1 | 93 | 2014-12-13 | 3 | True |
| 1 | 85 | 2015-01-19 | 4 | True |
| 1 | 91 | 2015-04-15 | 4 | True |
| 1 | 55 | 2015-04-27 | 3 | True |
| 1 | 59 | 2015-06-14 | 4 | True |
| 1 | 5 | 2015-06-14 | 1 | True |
| 1 | 83 | 2015-07-23 | 5 | True |
| 1 | 42 | 2015-08-22 | 5 | True |
| 2 | 120 | 2012-05-24 | 3 | False |
| 2 | 86 | 2012-07-29 | 5 | False |
| 2 | 49 | 2012-11-11 | 4 | False |
| 2 | 54 | 2012-12-07 | 4 | False |

**Table EC.2 The first review left for each open restaurant.**

| business id | user id | date | stars | open |
|---|---|---|---|---|
| 5 | 109 | 2009-08-05 | 4 | True |
| 3 | 75 | 2009-10-21 | 5 | True |
| 8 | 1 | 2011-08-17 | 4 | True |
| 1 | 8 | 2014-10-20 | 4 | True |
| 7 | 102 | 2015-04-23 | 5 | True |
| 6 | 52 | 2015-08-05 | 3 | True |

**Table EC.3 The last review left for each closed restaurant.**

| business id | user id | date | stars | open |
|---|---|---|---|---|
| 2 | 117 | 2013-10-12 | 5 | False |
| 4 | 124 | 2014-03-21 | 5 | False |

## EC.3. Supplementary numerical results

In this section, we provide additional results with respect to estimates of cuisine dummies in the customer choice model (Table EC.4), present examples of LFM predictions, first across a sample of cuisines for a specific zip code (Table EC.5) and second to compare several LFM performances including features (Table EC.6).

| Cuisine | Mean | S.E. | t value | P-Value | N |
|---|---|---|---|---|---|
| American | 1.786 | 0.075 | 23.867 | 0 | 513 |
| Asian | 2.084 | 0.097 | 21.533 | 0 | 114 |
| Bakeries | 1.289 | 0.312 | 4.139 | 0 | 44 |
| Bar | 2.057 | 0.107 | 19.136 | 0 | 103 |
| Barbeque | 1.783 | 0.158 | 11.316 | 0 | 104 |
| Breakfast | 2.303 | 0.081 | 28.482 | 0 | 141 |
| Buffet | 3.052 | 0.084 | 36.54 | 0 | 76 |
| Burger | 2.535 | 0.11 | 23.044 | 0 | 68 |
| Cafe | 1.587 | 0.17 | 9.326 | 0 | 92 |
| Chicken | 1.251 | 0.302 | 4.137 | 0 | 111 |
| Chinese | 1.313 | 0.189 | 6.956 | 0 | 227 |
| Delis | 0.319 | 0.689 | 0.463 | 0.644 | 77 |
| Diner | 2.021 | 0.202 | 9.989 | 0 | 37 |
| Eastern | 1.291 | 0.427 | 3.026 | 0.002 | 24 |
| Fast Food | 0.249 | 0.559 | 0.446 | 0.656 | 352 |
| Hawaiian | 1.426 | 0.284 | 5.019 | 0 | 40 |
| Hot Dogs | 0.766 | 0.553 | 1.386 | 0.166 | 60 |
| Indian | 1.99 | 0.197 | 10.093 | 0 | 35 |
| Italian | 1.595 | 0.119 | 13.351 | 0 | 241 |
| Japanese | 2.034 | 0.107 | 19.051 | 0 | 100 |
| Korean | 1.041 | 0.513 | 2.03 | 0.042 | 30 |
| Mediterranean | 1.576 | 0.212 | 7.449 | 0 | 57 |
| Mexican | 0.924 | 0.17 | 5.44 | 0 | 439 |
| Pizza | 1.543 | 0.148 | 10.416 | 0 | 240 |
| Sandwich | 1.267 | 0.233 | 5.444 | 0 | 115 |
| Seafood | 1.827 | 0.142 | 12.887 | 0 | 97 |
| Southern | 1.927 | 0.239 | 8.079 | 0 | 26 |
| Steak | 1.946 | 0.107 | 18.174 | 0 | 130 |
| Sushi | 1.944 | 0.096 | 20.262 | 0 | 147 |
| Thai | 1.605 | 0.167 | 9.617 | 0 | 92 |
| Vietnamese | 2.047 | 0.16 | 12.764 | 0 | 55 |

**Table EC.4 Estimates of cuisine dummies in the customer choice model.**

| **Cuisine** | **Actual Rating** | **Predicted Rating** | **Error-Squared** |
|---|---|---|---|
| American | 3.515 | 3.528 | $\approx 0$ |
| Asian | 4.250 | 4.031 | 0.047 |
| Bakeries | 4.000 | 4.036 | 0.001 |
| Barbeque | 3.583 | 3.597 | $\approx 0$ |
| Italian | 3.437 | 3.707 | 0.072 |
| Japanese | 4.250 | 3.846 | 0.163 |
| Mexican | 3.673 | 3.534 | 0.019 |
| Pizza | 3.875 | 3.229 | 0.416 |
| Steak | 3.954 | 3.840 | 0.013 |
| Sushi | 3.500 | 3.952 | 0.205 |
| **Test RMSE** | | 0.499 | |
| **Train RMSE** | | 0.410 | |

**Table EC.5 Comparison of the average predicted and actual ratings across a sample of cuisines for zip code** 89110

| Model ($R_i$) | Plain LFM $. \sim u_c^T v_z$ | LFM (Cuisine dummies) $. \sim \zeta_0 + \zeta_c c_i + \boldsymbol{u}_{c_i}^T \boldsymbol{v}_{z_i}$ | LFM (Cuisine dummies+features) $. \sim \zeta_0 + \zeta_c c_i + \boldsymbol{\zeta_y^T} y_i + \zeta_p p_i + \boldsymbol{u}_{c_i}^T \boldsymbol{v}_{z_i}$ |
|---|---|---|---|
| **Test RMSE** | 0.692 | 0.684 | 0.677 |
| **Train RMSE** | 0.645 | 0.644 | 0.639 |

**Table EC.6 Comparison of several LFM performances (RMSE) to predict ratings applied on test and training sets.**

### EC.3.1. Supplementary data analysis

In this empirical investigation, we present a detailed examination of *improved* restaurant locations, focusing on two distinct cuisine types—Asian and Mexican cuisines—within Las Vegas. The primary evaluation criterion in this study is the star ratings assigned to these establishments, serving as a pivotal indicator of their operational performance and customer appraisal.

Our study reveals a nuanced spatial distribution of highly-rated ($r_i > 4$) Asian and Mexican restaurants, notably delineating the unique cuisine preferences of Las Vegas. The South-West (SW) region, commonly denominated as Chinatown, emerges as a focal point for esteemed Asian establishments. Conversely, the North-East (NE) region showcases a concentration of highly-rated Mexican restaurants, with select establishments extending their influence into the SW part of the city (Figure EC.2 left).

To refine our assessment, we leverage the rich data provided by Yelp reviews, enabling the approximation of restaurant opening and closing dates. This temporal insight facilitates the calculation of survival probabilities.

Our findings, graphically depicted in Figure EC.2 (right), underscore a compelling narrative. Mexican restaurants in the NE region exhibit a significantly higher survival probability, while Asian restaurants demonstrate a greater likelihood of endurance in the SW region. This observation underscores a symbiotic relationship between cuisine type and *improved* geographical positioning, suggesting that strategic location decisions could substantially enhance survival rates by up to 37.5%.

Additionally, we compare the mean predicted demand for these cuisines across the delineated regions, considering their open/closed status. Table EC.7 reveals that surviving restaurants, on average, have a higher predicted demand than those that have closed. Furthermore, restaurants located in their most-favored regions tend to boast a higher average demand compared to their counterparts in alternative regions (i.e., Mexican, NE=30.99 versus SW=26.64). This example underscores the significance of incorporating local taste preferences into the location prediction model, emphasizing the impact on both predicted demand and overall restaurant survival rates.

| Mean Predicted Demand | Open | Closed |
|---|---|---|
| Asian SW | 31.56 | 28.60 |
| Mexican NE | 30.99 | 28.29 |
| Asian NE | 29.06 | 29.01 |
| Mexican SW | 26.64 | 24.54 |

**Table EC.7 Average predicted demand for specific cuisines in the alternative regions.**

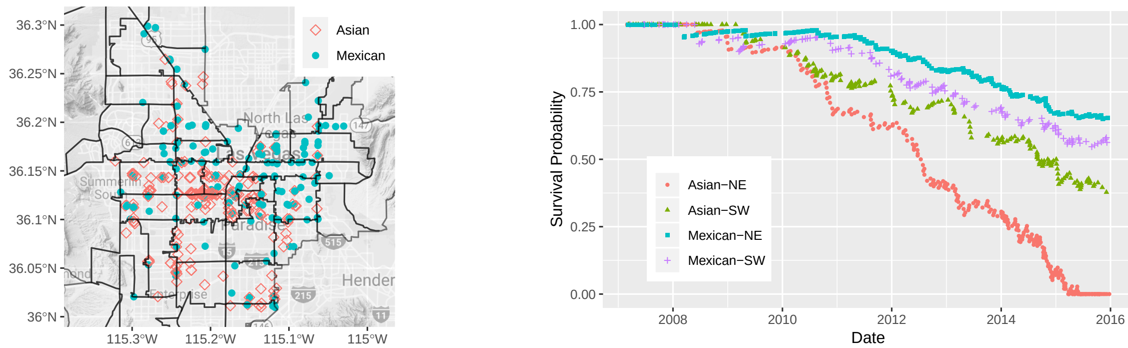


**Figure EC.2** **Most favored ($r_i \geq 4$) locations of Asian and Mexican cuisines and their survival probabilities.**

## EC.3.2. Sensitivity Analysis

In this part, we present sensitivity analysis as well as comparison with observed entry and exit patterns from the Yelp data.

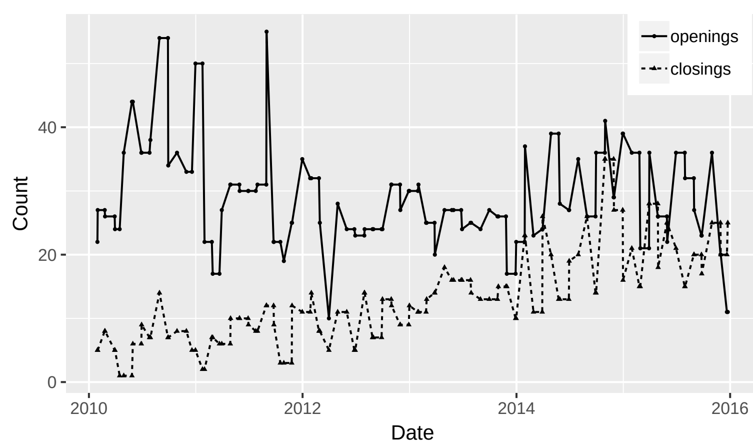


**Figure EC.3** **Restaurant openings and closings in observed data aggregated by month.**

Figure EC.3 shows restaurant openings and closings from the Yelp dataset aggregated on a monthly basis. While we do not know the exact dates, we rely on the initiation of reviews or the absence of further reviews as indicative signals to discern the dates of these entry and exit events from the review data.

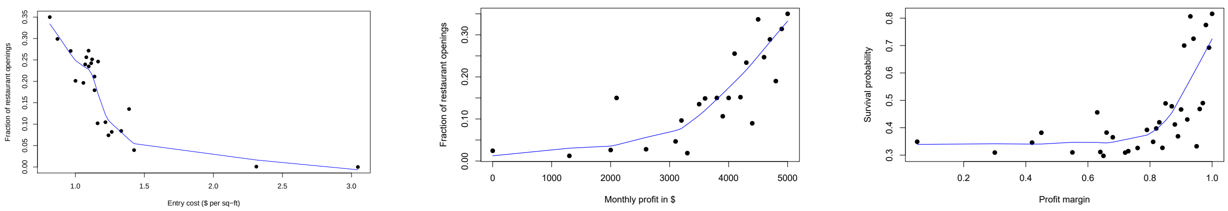


**Figure EC.4** **Sensitivity Analysis: For the similar price point ($ and $$) restaurants, fraction of restaurant openings per year versus entry cost, fraction of openings versus monthly profits and survival probability versus profit margins. Cost and profit calculations are based on our estimates and scaled.**

Secondly, we focus on a specific category of restaurants, specifically those with price points of $ and $$. We analyse the proportion of store openings per year, categorized by zip codes, in correlation with our estimates of entry costs. Upon examining Figure EC.4 (on the left), it is evident that an increase in entry costs results in a decrease in the number of restaurant openings within this particular category.

The second figure (on the right) within this illustration shows fractions surviving—derived from reviews data, as we infer closings from lack of reviews as mentioned at the beginning of this section—as a function of the profit margin (again our estimates). Notably, it reveals an escalating trend in survival probability with an increase in profit margin.

These visualizations from Yelp data not only underscore the volatility and complexity of the dynamics in location equilibria model but also offer insights into the sensitivity of restaurant openings to our estimates of entry costs and the consequential impact of our estimates of profit margins on survival probability.

## EC.4. Data sources and summary statistics

In this section, we provide sources of public datasets used in our empirical analysis and summary statistics of the so called competitor intelligence data (Tables EC.8 and EC.9).

| **Source** | **Information/Field** |
|---|---|
| Yelp Dataset Challange | Total number of reviews (proxy for demand), user rating distribution, restaurant location (lat., long.), average star rating of a restaurant, cuisine type, price point ($ to $$$$'s), |
| NASA, SEDAC (2015) | Grid point population count by lat-long., 0.927 km in height, 0.752 km in width |
| Factual API | Hotel location (lat. long.) and number of rooms |
| Google Maps API | Retrieves zip codes from lat-long. coordinates |
| OpenTable | Daily bookings for a sample of restaurants, used to correlate with number of reviews |
| Craigslist | Commercial rent prices at zip code level, ($/sqft) |
| United States Census Bureau American Fact Finder (2014) | Population: age, sex, race, households: total households, family households, income: household income, families, married-couple families, nonfamily households |
| Popular times feature, Google Maps | Restaurant occupancy in percentage |
| Restaurant seat and size (square footage) | City of Las Vegas Building Department |

**Table EC.8 A listing of all data sources used to estimate and validate our models.**

| Variable | Mean | Median | Std. Dev. | Max | Min | Count |
|---|---|---|---|---|---|---|
| *Yelp Restaurants* | | | | | | |
| Star Rating | 3.43 | 3.50 | 0.73 | 5 | 1 | 3,987 |
| Review Count | 132.21 | 42 | 297.19 | 5,642 | 3 | 3,987 |
| Distance from Grid points (avg.) | 5.32 | 5.01 | 2.1 | 14.62 | 2.39 | 1526 |
| *Demographics* | | | | | | |
| Local Pop. | 918.88 | 567.84 | 1023.48 | 7116.75 | 0 | 1526 |
| Tourist Pop. | 1446.99 | 485.1 | 2265.25 | 14515.52 | 44.1 | 250 |
| Income | 66,032.45 | 66,785.44 | 17,595.50 | 104,708 | 33,417.09 | 39 |
| Rent ($/sq ft) | 1.43 | 1.15 | 1.14 | 7.93 | 0.81 | 39 |
| *Popular Times Restaurants* | | | | | | |
| Capacity | 102.38 | 79.50 | 87.33 | 684 | 6 | 382 |
| Size (sq ft) | 1,166.26 | 1,000 | 661.41 | 4,500 | 50 | 382 |

**Table EC.9 Summary statistics of competitor intelligence data.**

Finally, we present summary information from our main data source, Yelp along with demographics. Figure EC.5 shows the type of information we can retrieve for a restaurant on Yelp's website. Tables EC.10 and EC.11 summarize the average number of restaurants and average review counts by price points and review star ratings of restaurants. In terms of distribution of reviews by star rating, Figure EC.6 reveals well-known J-shaped distribution. We also include the price-range convention used in Yelp.com (Figure EC.13) that is useful to convert the $ signs into monetary equivalences. In terms of demographics, we plot population density distribution of Las Vegas city

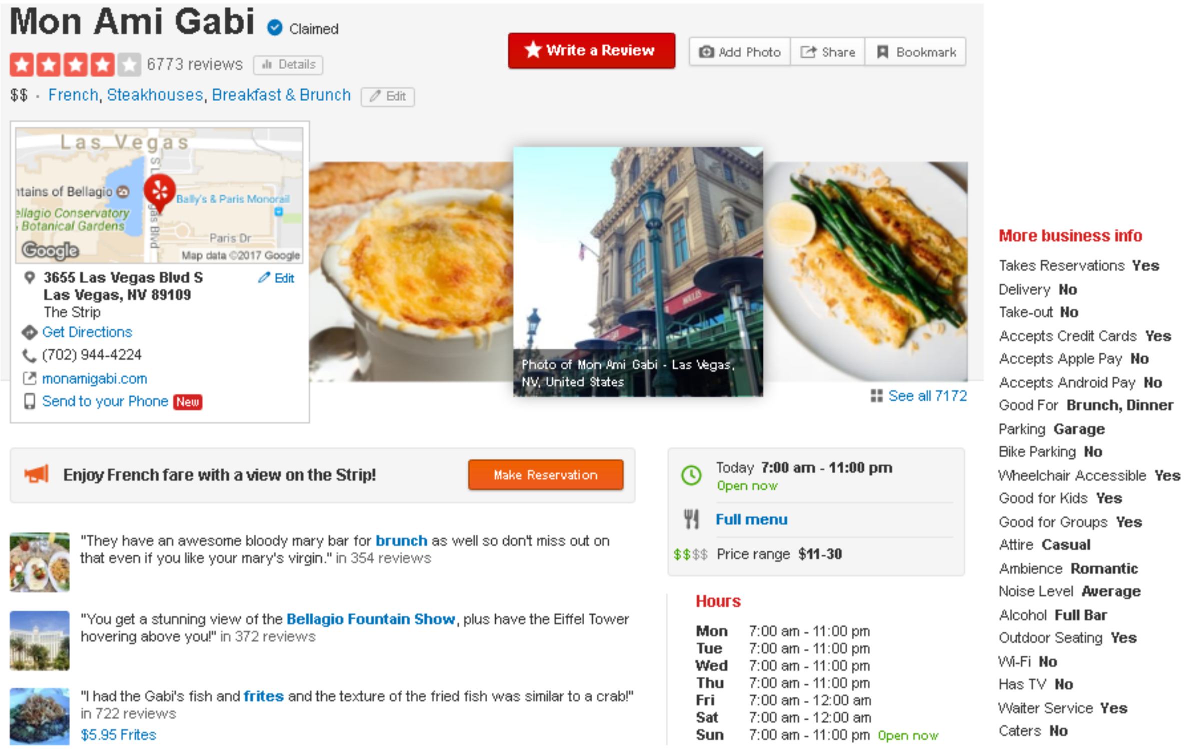


**Figure EC.5** **Example of a restaurant shown in Yelp's website in Las Vegas (U.S.).**

EC.7 and include income distribution for three income levels corresponding to low ($< 25K$), middle ($25K - 100K$) and high ($> 100K$) on Table EC.12.

| Price Point | Number of Restaurants | Average Review Count |
|---|---|---|
| 1 | 1532 | 64.80 |
| 2 | 1398 | 203.86 |
| 3 | 136 | 456.96 |
| 4 | 58 | 465.01 |

**Table EC.10** **Average number of restaurants and average review count by price point.**

| Stars | Number of Restaurants | Average Review Count |
|---|---|---|
| 1 | 11 | 7.54 |
| 2 | 222 | 28.52 |
| 3 | 913 | 73.54 |
| 4 | 1607 | 203.23 |
| 5 | 371 | 197.39 |

**Table EC.11** **Average number of restaurants and average review count by star rating.**

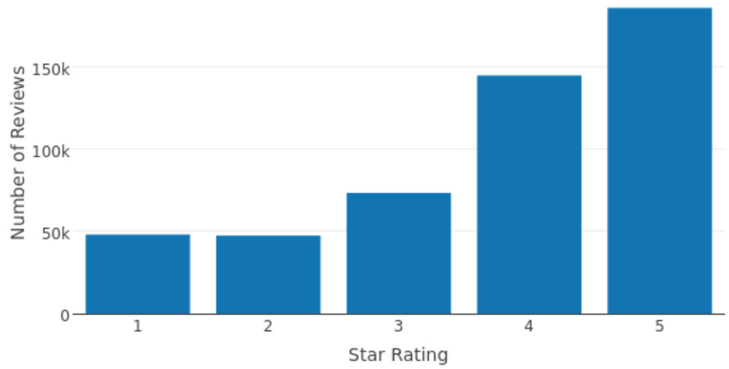


**Figure EC.6** **Distribution of reviews by star rating.**

| Income Range | Number of People |
|---|---|
| Low <25 K | 500818 |
| Middle 25 K – 100 K | 1239421 |
| High >100 K | 368573.1 |

**Table EC.12** **Income distribution of Las Vegas city.**

| Yelp Symbol | Actual Value |
|---|---|
| $ | $< 10$ |
| $$ | 11-30 |
| $$$ | 31-60 |
| $$$$ | $> 61$ |

**Table EC.13** **Price point convention used in Yelp.com.**

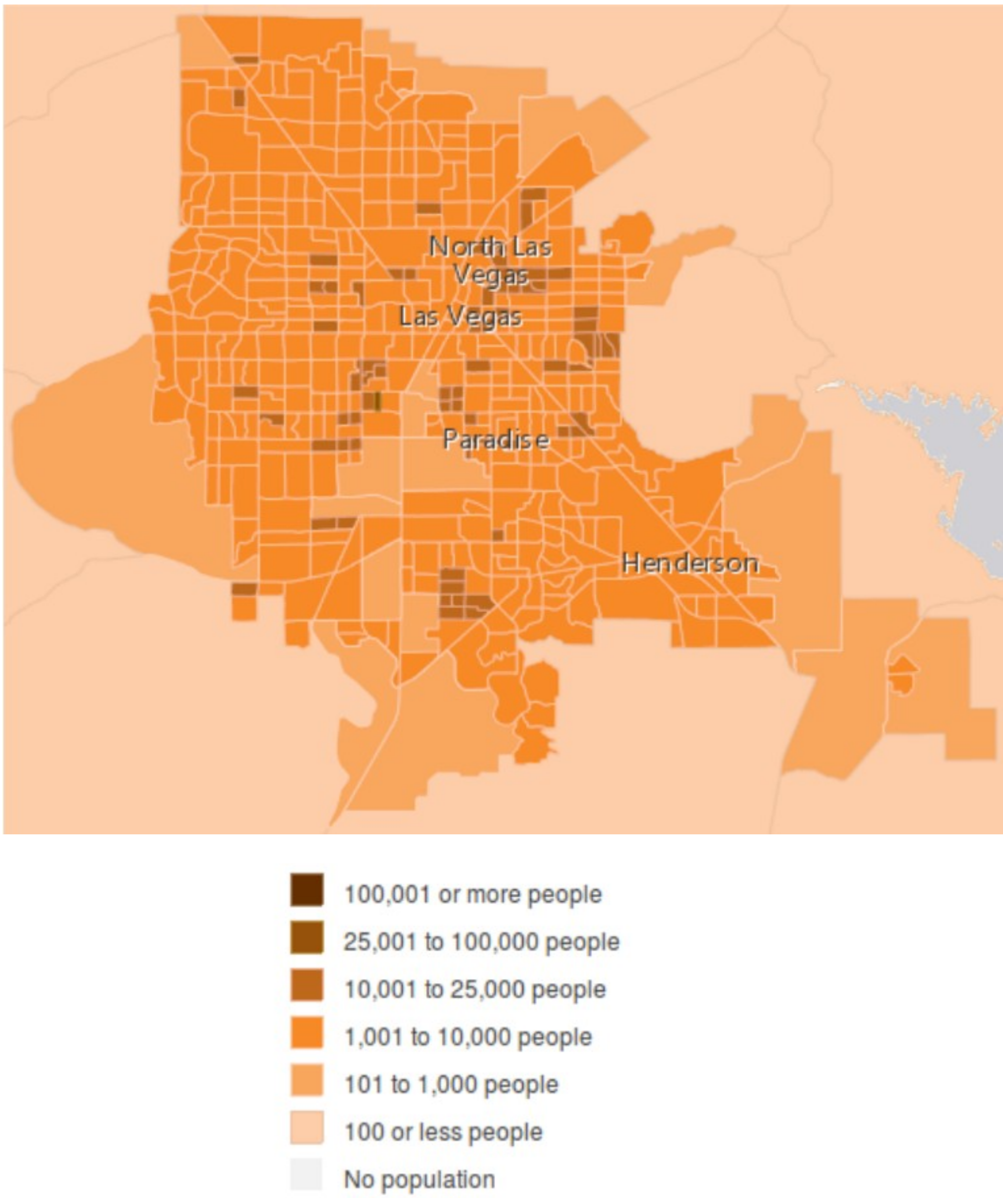


**Figure EC.7 Population density distribution of Las Vegas.**